September 9, 2026

The Trillion-Dollar Friction

A Structural Vulnerability in the U.S. Treasury Market

Toward Satisfying the Synthetic Duration Demand at the Source

Steven T. Williams[ς]

EXECUTIVE SUMMARY

A structural mismatch in the U.S. Treasury market, where over $1 trillion in institutional demand for synthetic duration is serviced through leveraged private-sector intermediation, creates a persistent source of systemic fragility, congests short-term funding markets, and imposes a compounding fiscal drag on the sovereign.

This paper demonstrates that the U.S. Treasury, as the natural producer of duration, possesses both the statutory authority and operational capacity to satisfy this structural demand directly at its source. We propose **Deferred Settlement Auctions** (DSAs), a procedural refinement to primary market architecture that allows long-horizon institutional investors to secure forward sovereign duration directly from the issuer.

The DSA framework is not a new derivative product, but an extended settlement term for primary debt issuance, featuring an automated monthly roll to complete the duration-delivery lifecycle in-house. Designed for seamless execution within existing sovereign debt management authorities and leveraging established Tri-Party custodial infrastructure, the architecture fully collateralizes counterparty risk and operates without altering the core price-discovery mechanics of the primary auction.

The framework delivers three principal dividends:

- **Mitigate Systemic Fragility:** De-leverages the Treasury ecosystem by providing an unlevered, direct channel for passive duration, substantially reducing reliance on repo-funded basis trades and mitigating March 2020-style fire-sale dynamics.
- **Stabilize Money Markets:** Relieves baseline demand in overnight Treasury repo markets, mitigating funding spikes, easing constraints on dealer balance sheets, and ensuring the smooth transmission of monetary policy.
- **Capture a Compound Fiscal Dividend:** Internalizes the embedded basis premium, tightens competitive auction clearing, and compresses risk premia across the yield curve. The framework yields a baseline **$1.7 billion** in annual interest savings, generating a 10-year Net Present Value of around **$15 billion**, with further upside as curve-wide risk premia compress.

* * *

Paper Structure: Part I establishes the strategic mandate for a sovereign solution and outlines its core principles. Part II delivers the complete operational blueprint. Part III validates the architecture and quantifies its economic impact. Part IV charts a prudent path to implementation via a pilot program.

[ς]Contact at steven.t.williams1@proton.me.
Project repository: treasury-dsa.github.io.

Part I

Foundations of a Sovereign Solution

I. Introduction

The U.S. Treasury market currently contends with a significant structural misalignment: a trillion-dollar divergence between how duration is issued and how it is ultimately accessed. While institutional mandates generate vast requirements for synthetic duration—interest rate exposure achieved through derivatives rather than cash bonds—the current market architecture forces this demand through a capital-intensive, basis-driven intermediation chain.

The resulting reliance on private-sector leverage to bridge the gap between primary supply and synthetic demand creates an inherent structural vulnerability. As evidenced during the market stress of March 2020, such an arrangement can transform stable institutional needs into sources of acute fragility when leveraged intermediation propagates recursive selling pressure.

While the baseline demand for synthetic duration is well-documented, a companion analysis highlighted that MBS convexity acts as a latent accelerant, generating non-linear surges in hedging flow that can overwhelm the private intermediation chain.[1] The persistence of such dormant risks in a systemically vital market underscores the danger of relying on highly levered, multi-tiered intermediation. This paper therefore moves beyond diagnosis to the design of a solution, shifting from a micro-analytic to a macro-policy lens.

We posit that a foundational solution requires addressing the mismatch at its source—with the end investors[2] and the ultimate issuer, rather than focusing on the private-sector intermediaries tasked with compensating for it. We argue that the U.S. Treasury, as the natural counterparty to the market's structural demand for duration, is uniquely positioned to resolve this inefficiency and neutralize the systemic risk it creates. The framework that follows provides the operational logic to close the architectural gap, enabling the sovereign to internalize the market's duration demand, enhance systemic resilience, and secure a direct fiscal dividend.

II. The Anatomy of a Structural Flaw

An effective solution requires analyzing the problem from first principles. The following analysis traces the market's core vulnerability from its origin, which is a trillion-dollar persistent demand for synthetic duration, through its underlying leverage mechanics, highlighting three primary areas of regulatory concern: financial stability, monetary policy transmission, and fiscal efficiency.

A. The Intermediation Engine: Mechanics of the Basis

While systemic risk crystallizes in the portfolios of hedge funds, its genesis lies not in their arbitrage strategies, but in the structural duration deficit that arises from the institutional strategy of traditional asset managers to overweight spread assets. This allocation approach, common among participants targeting outperformance, involves constructing portfolios of shorter-duration assets including Agency MBS, CLOs, short-tenor private credit,[3]

Acknowledgments: The author thanks James Aitken, Wenxin Du, Darrell Duffie, Alex Etra, Jay Kahn, Brad Setser, and Daleep Singh for helpful comments and discussions on earlier versions of this work, as well as several colleagues whose roles prevent public credit. All remaining errors, omissions, and policy views are solely the author's responsibility. Large language models were used for diagram generation, translation, and copyediting to ensure clarity and precision.

[1] Williams, S. T. (2024). "MBS Convexity & the Treasury Basis Nexus—Falling Rates, Rising Risks: Hedging Pressures of 4+ Cpn MBS."

[2] While the core drivers of significant synthetic duration exposures among end investors are clearly delineated, the interplay between these constituent forces is inherently complex. Current analysis attributes this demand primarily to a structural mismatch in desired duration between issuers and investors, persistent cyclical variations in credit appetite, or, balance sheet frictions induced by sovereign issuance patterns. For policy purposes, however, this etiological attribution is secondary; a demand of such scale and persistence creates architectural pressures regardless of its exact composition. Given that directly altering end-investor allocation is a far more draconian path to structural realignment, our investigation concentrates on the issuer side of the equation as the most effective lever for market enhancement.

[3] An emerging, if still nascent, accelerant of this gap is the vast allocation of private credit toward artificial intelligence and compute infrastructure. Because rapid hardware depreciation constrains such structured debt to short tenors, these allocations offer spread without duration, creating a prospective tailwind for synthetic overlays. Conceptually, this hints at a complementary macro equilibrium: private capital absorbing short-term technological and execution risks, while the sovereign supplies the underlying long-horizon duration anchor that naturally absorbs the eventual macro payoffs. Under prevailing market architecture, however, relying on leveraged

and other floating-rate assets. Such a construction creates a fundamental mismatch. It captures credit spread, yet leaves investors underallocated to duration relative to their governing constraints: long-term liabilities for pension funds and insurers, and index benchmarks for mutual funds.[4] To bridge this deficit, the most capital-efficient and liquid instrument is typically a CME-listed Treasury futures contract. Consequently, net end-user positions in these futures now consistently exceed $1 trillion,[5] reflecting a demand that is structural rather than tactical.

This sustained, one-sided pressure from end investors causes Treasury futures to trade at a small, persistent, and theoretically unwarranted price premium (the cash-futures basis), typically on the order of 3–8 basis points[6] relative to the underlying cash Treasury securities. The resulting pricing discrepancy creates a reliable arbitrage opportunity, primarily captured by hedge funds. The strategy is conceptually straightforward: sell the relatively expensive futures contract and simultaneously purchase a, usually cheap-to-deliver, cash Treasury bond to hedge the position. Because the basis is narrow, generating meaningful returns requires executing the trade at immense scale.

Achieving such scale requires substantial leverage,[7] readily provided by bank dealers through the repo market. The purchased Treasury bond serves as high-quality collateral, allowing arbitrageurs to secure the necessary overnight or short-term financing from money market funds and other cash lenders. Dealer accommodation of this leverage rests on the futures contract's physical settlement provision—a contractual delivery mechanism that guarantees basis convergence, provided positions are held to expiry.[8] This structural tether underpins the trade's perceived safety, encouraging intermediaries to finance massive gross positions against minimal haircut buffers. Figure 1 provides a schematic of this intermediated structure, illustrating the full sequence of transactions that connect end-investor demand to the cash Treasury market.

While the granularity of U.S. market data has fostered a clear consensus on these mechanics, the underlying dynamic is not a uniquely American phenomenon. Both the United Kingdom[9] and Canada[10] have observed a significant rise in similar basis trading activity, with emerging evidence pointing to the same root cause:[11] strong structural demand for synthetic duration from institutional end users. The systemic concerns this paper outlines should therefore be understood not as a local anomaly, but as exemplifying a foundational challenge endemic to the architecture of modern sovereign debt markets globally.

B. The Fault Lines: Systemic, Policy, and Fiscal

While functional under stable conditions, the current intermediation structure harbors significant latent stress. Four primary fault lines emerge from its design, creating vulnerabilities that, though masked during calm periods, threaten financial stability, compromise monetary policy, and impose a direct cost on the sovereign.

The first and most acute fault line is the systemic risk embedded in its core structure. A considerable fraction of Treasury securities—at times up to 6% of non-SOMA coupon issues—becomes tied to such repo-funded, leveraged positions. The involvement of multiple,

---

intermediaries to bridge this gap embeds systemic fragility into financial plumbing, while imposing a subtle friction on the buildout itself.

[4] Barth, D., Kahn, R. J., Monin, P., and Sokolinskiy, O. (2024). "Reaching for Duration and Leverage in the Treasury Market." Finance and Economics Discussion Series 2024-039. Washington: Board of Governors of the Federal Reserve System.

[5] Treasury Borrowing Advisory Committee. (2024). "Discussion of Treasury Futures Positions Across Different Investor Types." Charge prepared for the meeting of January 30, 2024. Washington, DC: U.S. Department of the Treasury.

[6] This figure represents the basis at the start of a quarterly contract cycle, expressed in price terms to highlight the direct arbitrage spread, a differential that is roughly four times smaller than its annualized yield equivalent.

[7] Zooming out from daily volatility to the underlying economics, the basis reflects the required hurdle rate of private capital. While leverage multiples in the trade (a simple metric distinct from the granular stress-loss models used for internal risk management) commonly range from 20x to 40x, and extend well beyond for specialized arbitrageurs, this gearing is not arbitrary. Consider a fund operating under a "2-and-20" fee structure. To deliver a net return of 8–10% to investors, the trade must generate a gross return on deployed capital of approximately 12–15%, a requirement intensified in multi-strategy platforms by the double-layered payout logic where PM-level cuts precede the fund's own incentive fee.

This creates a rigid mechanical trade-off. Even at 50x leverage, a 12% gross return requires the trade to offer an annualized spread (net of funding) of roughly 24 bps. If the spread compresses to 10 bps, the same return target necessitates increasing leverage to 120x, a level that is fundamentally misaligned with the market's capacity to absorb deleveraging during a crisis.

While specialized fixed-income funds may target lower net returns (e.g., 4–6%) and some participants margin the trade as a capital-efficient overlay within a single-counterparty, broader portfolio netting framework, the system as a whole remains trapped in an unappealing binary: a narrow basis requires extreme leverage, while a safer leverage profile requires a wider basis. Neither outcome is a sign of a fundamentally sound market architecture.

[8] In practice, the dominant execution is approximately CF-weighted rather than the 1:1 face-to-contract ratio implied by the delivery invoice formula. This weighting allows the participant to neutralize directional duration exposure and isolate the basis, which is essential for maximizing leverage capacity. A result of this choice is that the position cannot be delivered as-is; should physical settlement be pursued, restoring the 1:1 ratio at expiry requires squaring the delivery tail through additional secondary market transactions. We leave the mechanics of such execution aside, noting only that the alpha—and complexity—in these strategies is frequently found in the management of these very details.

[9] Bank of England (2025). "Financial Stability Report — December 2025." London: Bank of England.

[10] Uthemann, A. and Vala, R. (2024). "How Big Is Cash-Futures Basis Trading in Canada's Government Bond Market?." Staff Analytical Note 2024-16. Ottawa: Bank of Canada.

[11] Saporta, V. (2025). "The Evolving Liquidity Landscape." Speech, 7 November. London: Bank of England.

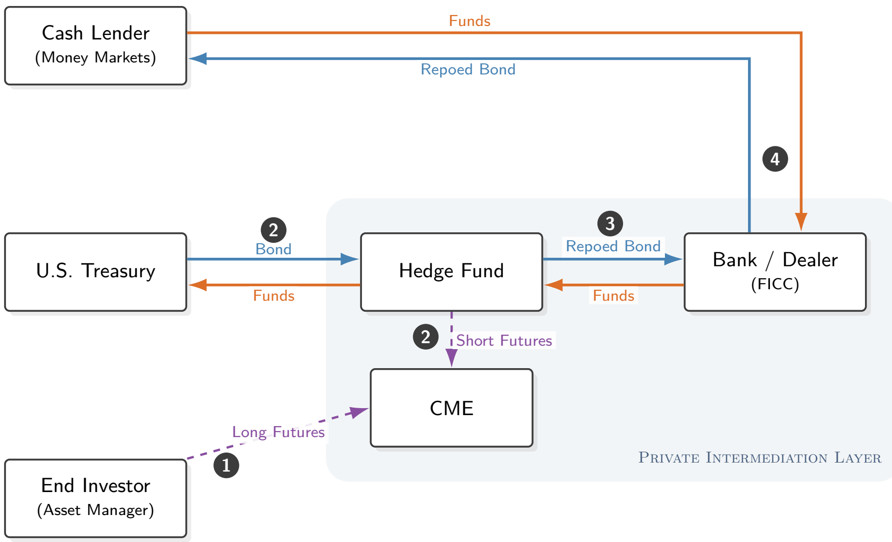

Fig. 1: The Intermediated Flow of Synthetic Duration. The diagram illustrates the multi-step, capital-intensive process through which structural demand for synthetic duration is currently met. The sequence is driven by a logical chain of incentives: 1 The process is initiated by an End-Investor's structural demand, which creates a premium in the Treasury futures market. 2 A leveraged arbitrageur (hedge fund) captures this premium by executing a two-part basis trade: simultaneously selling the rich futures contract while buying the underlying cash Treasury security. 3 The arbitrage is executed at scale by securing financing from a Bank/Dealer, which 4 in turn sources this liquidity from the ultimate Cash Lenders in the repo market, completing the funding circuit.

drawdown-sensitive intermediaries[12] creates a fragile chain; a shock[13] that forces leveraged participants to exit can trigger disruptive fire sales in the Treasury market.

This fire-sale dynamic reveals the second fault line: the acute burden placed on dealer capacity during crises. While hedge funds and principal trading firms facilitate market functioning in stable conditions, stress events shift the intermediation burden to bank dealers—the sole entities with direct access to Federal Reserve liquidity facilities. As Duffie et al. (2023) demonstrate, dealer capacity is not infinite; it is a finite balance sheet resource governed by a combination of post-GFC regulatory limits, internal profitability targets, and strict risk thresholds.[14] Crucially, the repo financing that dealers provide to the basis trade, even with the relief offered by sponsored repo, still consumes significant balance sheet capacity in aggregate. This absorption actively constrains dealers' ability to intermediate the Treasury market and, by extension, other critical markets, precisely when that capacity is most needed.

The third fault line runs through the funding markets themselves, impacting the transmission of monetary policy. The basis trade has migrated from *a* participant in Treasury repo funding to one of its primary organizing

[12] Intermediary retrenchment does not rely solely on lenders pulling repo financing. When volatility spikes, internal VaR limits, rising margin demands, and anticipated redemptions can force arbitrageurs to unwind positions regardless of funding conditions. For empirical evidence from March 2020, see: Kruttli, M. S., Monin, P. J., Petrasek, L., and Watugala, W. S. (2025). "LTCM Redux? Hedge fund Treasury trading, funding fragility, and risk constraints." Journal of Financial Economics 169.

[13] While the trigger for a basis unwind is often *exogenous*, the ensuing deleveraging is an *endogenous*, self-reinforcing spiral driven by the system's reliance on gross leverage. External pressure on the trade can be logically decomposed into two distinct channels. The breadth of potential actors involved suggests that such pressure is not an anomaly, but a recurrent feature of market stress.

On the *Cash Leg*, the pressure stems from the "dash for cash," where Treasuries are liquidated for reasons unrelated to their fundamental valuation. The roster of forced sellers is pervasive and systemic: it includes (1) liability-driven sellers, such as mutual funds meeting T+1 redemptions; (2) risk-driven sellers, such as volatility-targeting strategies deleveraging, leveraged funds facing margin calls on other assets, and pension funds or insurers defensively hoarding cash; and (3) banks liquidating HQLA to fund corporate credit line drawdowns. A critical, under-appreciated vulnerability here is (4), the degree to which U.S. financial stability remains susceptible to the liquidation of foreign reserves (whether from central banks conducting FX interventions or from SWFs meeting domestic fiscal needs), actions governed by factors exogenous to U.S. financial markets, ranging from domestic policy needs to broader geopolitical considerations.

The *Futures Leg* presents a less understood but equally potent vector, detailed in our companion analysis "MBS Convexity & the Treasury Basis Nexus." The mechanism is specific to the current regime: asset managers now hold a historic concentration of high-coupon, negatively convex Agency MBS. A sharp decline in benchmark yields accelerates prepayments, causing the duration of these portfolios to mechanically collapse (a convexity, or rather $\mathrm{d}D/\mathrm{d}r$, gap of nearly 3x the index). This forces managers to buy Treasury futures in massive size, estimated at \$140 bn for a 100 bps move, to maintain benchmark compliance. While the fundamental repricing in a crisis often pushes the entire rates complex lower, microstructure dynamics present the primary tail risk. In an acute stress scenario, the rally that activates mechanical futures buying from MBS hedgers would coincide with a liquidity squeeze forcing non-fundamental selling in the cash market. The key issue is less the absolute level of rates than the potential for severe dislocation in their relative pricing. Compounding pressures would widen the basis beyond intermediary capacity, triggering a disorderly deleveraging cascade.

[14] Duffie, D., Fleming, M. J., Keane, F. M., Nelson, C., Shachar, O., and Van Tassel, P. (2023). "Dealer Capacity and U.S. Treasury Market Functionality." Staff Reports 1070. New York: Federal Reserve Bank of New York. See also Du, W., Hébert, B., and Li, W. (2023). "Intermediary Balance Sheets and the Treasury Yield Curve." Journal of Financial Economics, 150(3), 103722, for an empirical term-structure framework modeling the yield-curve impact of post-GFC dealer balance-sheet constraints.

drivers, now accounting for up to a third of all daily Treasury repo transaction volume.[15] With a net absorption of lender cash that vastly exceeds its gross trading share, the trade exerts a direct price impact on the short-term rate complex. Because these positions are governed by the mechanical imperatives of arbitrage[16] rather than outright rate sensitivity, repo demand is profoundly inelastic.[17] Consequently, any marginal contraction in lending supply is felt acutely in funding rates.

The dynamic crystallized as liquidity buffers depleted, with Q3-2024 quarter-end pressures serving as a precursor to the binding reserve regime of 2025. In this environment, the basis trade acts as a systemic bottleneck. While reserve-neutral in aggregate, it saturates MMF repo capacity and monopolizes the dealer balance sheets required for circulation. The disproportionate SOFR spikes triggered by routine swings in the Federal Reserve's non-reserve liabilities, and the resulting compulsion of policymakers to explore new asset purchases, underscore a sobering reality: monetary policy transmission has become unnecessarily encumbered by the market's underlying mechanics.

Recent research from the Dallas Fed offers an initial empirical foundation for these dynamics, suggesting that the sensitivity of tri-party general collateral rates to leveraged repo borrowing has lately been nearly double its sensitivity to the level of reserves.[18] This indicates that in recent market regimes, the net funding demand of the basis trade has often challenged aggregate liquidity as the more immediate determinant of the market's anchor rates.

Inevitably, this dynamic has repercussions for the ongoing debate over the Federal Reserve's long-term balance sheet normalization and the banking system's lowest comfortable level of reserves (LCLoR). Beyond standard HQLA/LCR requirements and discount window stigma, focus has increasingly turned to the intraday distribution of liquidity. While empirical research has focused on Fedwire payment delays, market color points toward a similar, albeit less quantified, dynamic in the repo complex: the morning execution of DVP-settled arbitrage positions often outpaces the finalization of (tri-party initiated) funding from money market lenders. Navigating this transient gap in funding certainty compels bank holding companies to maintain elevated precautionary reserve buffers, further locking up systemic liquidity.

The transition toward balance sheet normalization is constrained by these structural rigidities.[19] Regardless of the specific mechanism used to optimize reserve demand, determining the system's true liquidity floor remains challenging; without more elastic plumbing, attempts to further reduce reserves risk inducing sudden and disproportionate funding volatility.

The final fault line is the clear fiscal signal sent by the prevailing architectural mismatch. The persistent richness of the futures basis, along with the negativity of long-end swap spreads, is unequivocal evidence of a mismatch between how the Treasury sells debt and how the market prefers to consume it. That results in two primary vectors of fiscal inefficiency.

The first is the direct dissipation of value: the premium investors pay for synthetic duration currently accrues to the intermediary chain rather than the sovereign issuer. However, this is compounded by a broader degradation of the sovereign's issuance and risk profile. By forcing duration through a leveraged pathway, the architecture compromises the efficiency of primary price discovery, embedding latent fire-sale risk into the security. Such instability erodes the bond's dual function as an essential hedge: it introduces a "wrong-way" price correlation, causing sovereign debt to sell off in concert with risk assets, while simultaneously degrading its deep liquidity precisely when that liquidity is most valued. Such latent instability effectively elevates the sovereign's borrowing cost across the entire curve to account for this contingent risk.

C. *The Core Diagnostic Question*

This confluence of pressures, spanning systemic stability, funding market function, and fiscal efficiency, raises a critical diagnostic question: are these distinct sources of fragility to be managed independently, or are they symptoms of a single, deeper flaw in the market's architecture? The answer determines the appropriate scope of any remedy. Yet, defining that scope requires identifying why these acute fault lines are currently tolerated as an inescapable feature of the status quo.

Such logic rests on a tacit but entrenched consensus: that the sovereign must, for want of a direct alternative,

[15] Derived on a look-through basis: each dollar of basis-trade repo borrowing generates a corresponding dealer transaction with a cash provider, creating two distinct repo contracts per economic position. Swap spread trades, structurally identical in their intermediation chain, would increase the estimate further. By this measure, basis-type trades account for virtually all Treasury repo market growth over the preceding three years.

[16] The frequent use of term-repo or duration-hedged financing further immunizes such positions against overnight rate volatility, effectively hardening the trade's demand curve.

[17] Ramaswamy, S., De Vere, H., McCormick, M., and Searls, S. (2025). "How sensitive is the Treasury cash-futures basis trade to funding condition shifts?" Federal Reserve Bank of Dallas.

[18] Kahn, R. J. and McCormick, M. (2026). "Rising hedge fund leverage affects monetary policy implementation." Federal Reserve Bank of Dallas.

[19] There is an interesting broader dynamic at work: by suppressing term premia, large central bank balance sheets may encourage the very shift into spread assets that creates the duration deficit in the first place. Structural improvements that allow balance sheet normalization to proceed safely would help ease this upstream pressure, even if the fundamental appeal of credit spreads ensures the bulk of structural duration demand persists.

remain an indirect participant in an intermediation chain whose resilience it can neither guarantee nor, given the scale of issuance, allow to fail. It accepts the view that, as the historical base of price-insensitive buyers has receded, leveraged intermediation is now a functional prerequisite for clearing the cash market.[20] This presumed dependency rests on a conflation of transactional success with architectural resilience; or, to be less charitable: *just because the bonds get sold doesn't mean the system works.* To assume the prevailing leveraged pathway is the only possible reality is to mistake a habituated workaround for a structural necessity.

* * *

Before defining the characteristics of an ideal solution, we must first confront the market's current trajectory. The regulatory and market consensus has coalesced around central clearing as the primary mechanism to modernize the market's architecture. Yet, to determine if this modernization is sufficient, we must subject it to a precise functional audit. Does the transition to cleared markets resolve the core diagnostic question, or does it merely shift the locus of the stress? By establishing the boundary conditions of what clearing can—and cannot—achieve, we define the necessary scope for a genuine architectural solution.

III. The Limits of the Clearing Mandate: Why Netting is Not Elasticity

In discussing the evolution of Treasury market structure, and as we turn toward durable solutions, there is no way around central clearing. Its implementation provides significant benefits addressing long-standing foundational vulnerabilities: it facilitates multilateral netting to enhance balance sheet efficiency, eliminates the opacity of bilateral leverage, prevents the propagation of settlement failures, and standardizes risk management across the ecosystem. These are vital upgrades to the market's plumbing that enhance its integrity and lay the necessary groundwork for market evolution, including the potential for scalable all-to-all trading.

However, it is important to delineate the scope of this solution. While the consensus rightly celebrates the credit hygiene provided by clearing, it stops short of guaranteeing liquidity resilience. The mandate is designed to sanitize the settlement chain, not to manufacture risk appetite. Consequently, the reliance on clearing as the primary remedy for basis trade risks rests on a heavy assumption: that the multilateral netting of normal flows will somehow liberate enough dealer capacity to absorb a gross notional liquidation during stress. Such logic risks equating the technical compression of risk on a balance sheet with the creation of the economic elasticity required to catch a falling knife.

The urgency of this distinction is underscored by the rigid capital constraints that govern the basis trade. Operating on arbitrage-thin spreads, the trade's viability is entirely contingent on capital efficiency; should collateral requirements be imposed without recognizing offsetting positions,[21] the return on committed capital would fall below arbitrageurs' hurdle rates. To restore equilibrium, the basis would have to widen materially, significantly increasing the cost of hedging for the end user. To avert this outcome and maintain essential liquidity, the current implementation trajectory is converging on a regime of portfolio-based cross-margining, where the CCP recognizes the hedge between cash and futures to minimize collateral obligations.

This path presents a fundamental paradox. Because sophisticated participants already operate near the efficient frontier of leverage via consolidated fixed-income financing, the transition to central clearing offers limited incremental benefits to basis pricing or intermediation capacity.[22] Consequently, the move to correlation-based

[20] Beyond primary auction clearing, this intermediation is frequently perceived to be associated with the provision of secondary market liquidity for off-the-run securities, yet such utility is structurally constrained by the CME's rigid delivery architecture. A Treasury security is excluded from delivery baskets for the vast majority of its existence; a 10-year note, for instance, spends about 80% of its life—including nearly its entire deeply off-the-run phase—outside any delivery window. The adoption of 'Ultra' contracts further intensifies this structural transience.

While non-CTD or non-eligible securities can be traded against futures, the absence of a contractual delivery anchor introduces idiosyncratic spread risks that preclude the enhanced capital efficiency (and resulting scale) characteristic of the primary basis complex.

Consequently, the basis trade provides fragmented, episodic liquidity rather than continuous market depth, anchoring itself squarely to the CTD boundary and leaving the rest of the off-the-run curve to be cleared by lower-leverage, relative-value buyers.

[21] Effective risk management requires that capital charges reflect the economic reality of the holistic position architecture, not the optical risk of isolated components. As analyzed by Kahn & McCormick (2025), the imposition of uniform minimum haircut floors on individual repo legs often conflates low margins with laxity. In the context of a hedged portfolio, a zero or even negative haircut can represent a rational, "proportionate" assessment of the net exposure. To enforce arbitrary floors on specific legs without recognizing the broader structural hedge is to impose a liquidity tax that degrades market capacity without a commensurate improvement in safety.

[22] It is a misconception that the basis trade typically operates far from the efficiency frontier of capital that central clearing promises to unlock, or that low bilateral haircuts are ipso facto evidence of lax risk management.

While inter-CCP offsets have been a structural reality for dealer house accounts for some time, their expansion to end-user clients has historically been delayed by the operational and legal frictions inherent in bridging the U.S. Treasury and futures complexes. The current emergence of formal customer conduits largely represents the institutionalization of the internal netting and risk-modeling sets that have long allowed sophisticated participants to closely cross-margining does not delever the system so much as commoditize its leverage. It extends the frontier's capital-efficient architecture to participants lacking the institutional discipline to manage it, effectively democratizing the risk while concentrating over a trillion dollars of exposure into a single node.

The shift introduces a distinct structural trade-off: we are replacing a distributed network of bilateral risk—which is opaque to regulators but retains the capacity for forbearance or cross-product pledges based on a dealer's firm-wide insight into the client's capital position—with a centralized architecture defined by algorithmic rigidity. While a CCP eliminates daisy-chain risk, it is bound by rulebooks to issue immediate, procyclical margin calls. This transforms the nature of the fragility from one of credit contagion to one of synchronous liquidation, creating a centralized pressure point that acts as a systemic accelerant when its models are breached, leaving no non-sovereign line of defense.

In such a scenario, the CCP ensures the *legal* default process is orderly, but it is powerless to manage the *economic* shock. As the clearinghouse attempts to auction hundreds of billions in gross notional collateral, it faces a market where dealer balance sheets are already constrained by the same volatility, which simultaneously triggers procyclically contracting internal risk limits, inflates risk-weighted assets (RWA) from client guarantees, and threatens binding G-SIB surcharge thresholds. In this environment, there is no natural buyer. The result is not a market-clearing price, but a liquidity vacuum. The transparency provided by the CCP becomes merely a window into a systemic seizure that no amount of margin can cure.[23]

* * *

Ultimately, this analysis is not a critique of central clearing, but a delineation of its natural scope. While the mandate, if well calibrated and meticulously implemented, provides the essential credit hygiene required to modernize the market's infrastructure, it is widely recognized that clearing operates as a credit-standardization tool rather than a liquidity-provisioning tool. It does not resolve the fundamental constraints on principal risk transfer—the very friction rendering the basis trade such a uniquely fragile systemic vulnerability. To burden the clearing ecosystem with the expectation of absorbing a liquidation that outstrips the market's unconditional liquidity is to demand physics from plumbing it cannot deliver. Indeed, the long-term resilience of the central clearing project depends on ensuring this critical infrastructure is not structurally overloaded with static duration risks threatening its solvency during a correlation break.

A durable resolution requires a companion architecture: a mechanism that works alongside the private intermediation layer to relieve it of the asymmetric structural demand, specifically the immense volume of asset manager futures buying currently serviced primarily by leveraged arbitrage. Only by resolving this asymmetry can the system be secured against the gross reality of a crisis that no amount of netting can conceal.

IV. Characteristics of a Good Solution

Effective solution design hinges on a clear articulation of desirable characteristics. For the purposes of this analysis, we posit the following criteria:

1) **Deleveraging Focus:** The ultimate effectiveness of any solution rests on its ability to structurally reduce the system's reliance on private sector leverage as the primary balancing mechanism.

2) **Futures Market Integrity:** The solution must acknowledge and protect the indispensable role of the futures market in providing market-wide liquidity and real-time price discovery.

3) **Preservation of Yield Compression:** Mechanisms should sustain the yield-reducing benefits derived from hedge fund sovereign debt absorption and end-investor spread asset allocation.

approximate the economic equivalent of cross-margining.

Such de facto efficiency is achieved through a standard, albeit bespoke, market practice: by consolidating a basis book with a single dealer, the repo leg and the futures leg are risk-managed as a unitary package. This typically utilizes dual-model calculations that contrast mandated regulatory margins with a dealer's own internal risk assessment. While the initial margin at the futures exchange is contractual and irreducible—and typically grossly disproportionate to the net risk of a hedged position—dealers leverage a holistic view of the position's offsetting PnL characteristics. Recognizing the package's net stability, they routinely set repo haircuts to near-zero, at times extending credit to fund the gap between regulatory margin requirements and their own 'house' risk views. Where competitive pressures dictate, they may even absorb the carry costs of this capital out of their own funding book. Furthermore, the repo leg is increasingly executed via Sponsored Repo. This allows the dealer to net the balance sheet impact (SLR), effectively trading a leverage-ratio constraint for a risk-based capital requirement derived from the dealer's role as the client's credit guarantor. Thus, the capital efficiency regulators seek to introduce already largely serves as the operative baseline for the current market reality.

[23] While the concentration of risk is an inherent feature of central clearing, the specific inclusion of the highly leveraged basis trade introduces a unique convexity profile. Unlike outright cash trades, the basis trade is a correlation wager susceptible to rapid, non-linear dislocations that traditional VaR models struggle to capture. By aggregating \$1 trillion of jump-to-dislocation risk, the system risks exhausting the CCP's default waterfall in a single tail event, endangering the CCP's ability to perform its core utility functions—clearing cash and repo—precisely when they are most needed. Such an impairment would radiate far beyond the Treasury complex, as a failure at the system's "taproot" would instantly compromise the collateral integrity of all other cross-asset clearinghouses (see Yadav & Younger, 2025). Furthermore, because the private sector cannot absorb a liquidation of this magnitude, the architecture effectively institutionalizes the sovereign backstop, rendering Federal Reserve support less a break-glass option than a requisite component of the default waterfall.

4) **Fiscal Prudence:** Ideally, measures should be self-financing, avoiding undue fiscal burdens.

5) **Market-Driven Adoption:** Proposed solutions should be incentive-compatible and endogenously superior, driving voluntary adoption through demonstrable advantages, not through regulatory mandate or the suppression of alternatives.

Underlying these characteristics is the requirement for **architectural integrity**. A solution's design should be internally consistent and demonstrably more resilient than the structure it seeks to replace, ensuring all components work coherently to resolve known points of failure and provide systemic clarity of function.

V. Potential Solutions

Have solutions already been proposed? Yes. While a critical review of these follows,[24] readers wishing to proceed directly to this paper's core proposal may turn to Section VI: The Natural Counterparty.

A. *The Appeal (and Drawbacks) of end-investor repo*

End-investor repo is often proposed as a solution, funding additional purchases of government bonds to provide added duration, thus obviating the reliance on futures. While many institutions (bond funds, SMAs, insurers, pension funds) possess some repo capacity, limited practical application at the end-investor level suggests inherent challenges.

First, a crucial accounting discrepancy for '40 Act funds exists under financial reporting standards mandated by SEC Form N-1A, where interest expense from repo financing flows into a fund's published expense ratio, while the economically equivalent financing cost embedded in a futures contract does not. This creates a strong incentive to favor futures, a bias amplified by data vendors whose widely followed expense metrics incorporate repo costs. The friction is compounded by a balance sheet asymmetry; repos are structured as on-balance-sheet borrowing that expands gross leverage ratios, whereas futures are derivative contracts managed off-balance sheet.

Futures markets, conversely, offer an efficient solution for large transactions: fully electronic execution, low fees, pre-trade price transparency, and contractual security—benefits inherent to a cleared central limit order book. That operational architecture directly addresses long-term investors' needs, offering operational efficiency unmatched by repo (and swap) markets.

Is end-investor repo a panacea? Partially. It addresses certain criteria, specifically **(2)** and **(3)**, for a good solution. Regarding **(1)**, it would indeed lessen the reliance on hedge fund arbitrage, potentially enhancing market stability during periods of stress. However, the fundamental dependency on repo funding persists. Should money market funds, or similar lenders, become risk-averse and restrict lending, any framework predicated on substantial repo volume would face systemic vulnerability.

The pivotal question, however, remains: Will end investors with duration targets realistically seek to replicate the complex, OTC-centric, and counterparty-intensive infrastructure essential for efficient repo financing at scale? Beyond banks, this capability is concentrated within a limited number of hedge funds. While not insurmountable, the requisite expertise is a constrained resource. Absent such knowledge, and lacking prioritized counterparty relationships, less sophisticated participants should anticipate material adverse selection costs, particularly in the initial phase.

B. *Swaps*

A first glance at the objective, reducing asset managers' large Treasury futures longs, suggests a shift to interest rate swaps. Some advocate such a move, echoing overlays historically associated with certain prominent Newport funds. However, an industry-wide swap push is flawed. SOFR swap spreads are generally no better than the cash-futures basis, and have been empirically much pricier at the long end of the curve post-GFC. Futures' quarterly convergence via physical delivery is key. It offers arbitrageurs greater leverage than with swap spreads, where the spread exposure carries the full duration of the tenor rather than just the time-to-expiry. Because full-tenor duration generates a vastly higher spread DV01 per unit of notional, risk limits force arbitrageurs to run much smaller aggregate positions, severely limiting the trade's scalability.

Banks can do some short-tenor swap arbitrage from an HTM perspective. However, broad end-investor demand for swaps would likely overwhelm bank capacity. The entire spread curve would be pushed toward the 30-year tenor. With long-end swap spreads already well below -50 basis points, these levels are unsustainable for most end

[24] Distinct from the proposals evaluated in this section, other major initiatives focus on easing constraints within the dealer-centric ecosystem. These include potential relief on the Supplementary Leverage Ratio (SLR) to unlock balance sheet space, and the broadening of the Standing Repo Facility (SRF), specifically via improvements to timing, clearing integration and counterparty eligibility, to ensure a robust liquidity backstop. While these reforms effectively increase the theoretical bandwidth of the intermediation sector, they do not surgically address the root cause of the congestion: the massive, one-way structural demand for synthetic duration that currently has no direct outlet.

investors.

Similar issues apply to other bilateral solutions (bond forwards, total return swaps, etc.). Even if sometimes cheaper now, use at scale reveals the same constraints. Absent offsetting client flows, intermediaries would still be forced to warehouse repo-funded Treasuries, failing to reduce repo reliance. Indeed, the persistent richness observed in the USD SOFR swap market stems from the same pervasive end-investor need for synthetic duration seen in the futures market; a supply solution designed to directly satisfy this underlying investor demand could consequently be expected to ease such imbalances across both markets.

C. *The Limitations of Increased Bill Issuance*

Some commentators (e.g., Kelly, 2024)[25] suggest the basis trade could be eliminated by significantly increasing Treasury bill issuance. The thesis is that insufficient T-bill supply creates latent demand for short-term safe assets, providing the "funding ground" for the trade. Increasing bill supply, it is argued, would both crowd out repo financing and reduce futures demand by lowering the duration of key benchmarks that asset managers follow.

This approach, however, rests on a conception of both funding dynamics and institutional demand that appears inconsistent with observed market structure. While the proximate cause of the cash-futures spread is not in dispute, attributing the trade's scale to a scarcity of money-like assets is questionable. A true surplus of cash seeking safe assets would depress bill and repo rates;[26] such a condition cannot, however, account for the creation of the profitable arbitrage spread that attracts arbitrageurs. The market's ability to finance the trade instead reflects the elasticity of modern funding markets to accommodate any profitable, collateralized opportunity.[27] Under this framework, funding follows opportunity; it does not precede it.

More critically, the assumption that futures demand is primarily a function of benchmark duration is inconsistent with observed investor behavior. Market benchmarks are not exogenous drivers but are themselves shaped by the interplay of fundamental investor demand and sovereign supply. The demand for futures is more structural and inelastic than this model suggests, stemming from at least two powerful, non-benchmark-related sources:

- **Liability-Driven Imperatives:** An investor with long-term liabilities must achieve a specific duration target. If preferred private sector credit assets have a shorter duration (which is common, as few non-sovereign borrowers can issue favorably beyond ten years), futures (or swaps) are used to bridge the resulting duration mismatch. This behavior is driven by fundamental risk management, not benchmark tracking. Indeed, sophisticated investors will adopt custom benchmarks if standard indices cease to reflect their liability structures.
- **Capital Efficiency and Leverage:** For many investors, futures are a first choice, not a last resort. Gaining duration exposure synthetically is vastly more capital-efficient, allowing physical cash to be deployed into higher-return strategies. This strategic demand for leverage can be cyclical and exists independently of government issuance patterns.

The proposed solution therefore addresses a symptom, the repo financing, instead of the underlying cause.[28] It would do little to meaningfully impact the duration

[25] Kelly, S. (2024). "Can the Treasury Kill the Basis Trade? (Gently, of course)." Without Warning Research.

[26] Extending the "crowding out" logic to its functional limit reveals the specific, and potentially destabilizing, market frictions necessary for bill supply to actively suppress the arbitrage incentive. Taken literally, a surge in bill issuance primarily induces a parallel upward shift in the short-term rate complex, an effect akin to a standard interest rate increase. This raises rates on bills, repo, and all instruments priced via expectations (including notes, bonds, the implied financing costs of futures) and the short-tenor spread products held by asset managers. Because the spread between implied and actual financing remains unchanged, the fundamental incentives for both the asset manager's synthetic overlay and the arbitrageur's basis trade are preserved. The trade only faces pressure if the repo-bill spread widens, a condition contingent upon a conscious saturation of dealer balance sheets or a failure in cash recirculation. That dynamic would effectively entail the destabilization of the repo market, a result standing in direct opposition to the Federal Reserve's stability mandates. Only then, with repo rates elevated relative to bills, would hedge funds be forced to demand richer futures prices to maintain the net basis. It is only at this point, by rendering the synthetic duration overlay more expensive for the end investor on price grounds, that the trade would be curtailed as demand for the hedge evaporates and the arbitrageur consequently loses their counterparty.

[27] We distinguish here between the static *stock* of cash available for lending and the *circuit* through which it flows. In the basis trade, financing is not a one-way transfer. Cash lent by money market funds to arbitrageurs is used to purchase Treasury securities; these proceeds are then typically redeposited by the bond seller back into money markets. This high-velocity cash circuit allows a finite stock of lender cash to support a much larger notional of repo-funded positions. The binding constraint is therefore neither the aggregate stock of base money nor a static pool of savings, but the cost of the net balance sheet footprint of the repo leg—a factor that has, in practice, proven highly elastic due to netting efficiencies. As such, the absolute stock of T-bills is a secondary factor.

[28] We can formalize these objectives to see why multiple pathways to a synthetic duration overhang exist, of which the one suggested is only a narrow subset.

Institutional investors solve a constrained maximization problem, seeking to maximize alpha (α) from credit spreads (CS) and sovereign term premia (TP), subject to a primary duration target (D_{target}), often derived from their liabilities:

$$\max \mathbb{E}[\alpha(CS, TP)]$$

Conversely, Treasury solves a minimization problem, seeking to reduce its expected long-term borrowing costs by issuing at tenors (T) where the term premia ($\mathbb{E}[TP(T)]$) are most favorable:

$$\min_{T} \mathbb{E}[\text{Cost}(\text{Issuance})]$$

The resulting supply of debt, which emerges from the intersection of these fundamental demand and supply curves, in turn shapes the of major benchmarks like the Bloomberg US Aggregate (which excludes securities with less than one year to maturity) and, more importantly, would not address the structural and strategic demand for futures. The likely outcome would not be the 'death' of the basis trade, but a more fragile market. Unabated demand for futures would chase a smaller supply of deliverable bonds, while increased bill issuance would simultaneously raise funding costs, constraining the very arbitrage needed to contain pricing dislocations. The core issue is not an undersupply of money-like assets, but a legitimate and structural need for synthetic duration that requires a more direct and resilient fulfillment mechanism.

D. Federal Reserve Purchases of Basis Packages in Crisis

A recent proposal by Kashyap et al. (2025) for managing acute market stress considers direct Federal Reserve intervention.[29] Their framework proposes that the Fed, during periods of significant market stress, could purchase entire cash-futures basis packages: simultaneously acquiring cash Treasuries and selling corresponding Treasury futures.[30]

Such direct absorption of positions being unwound by leveraged participants would likely prove highly effective in stabilizing markets, offering particular utility if stress also emanates from the futures side (as detailed in our companion paper). Furthermore, as the authors highlight, fully hedged positions ensure these interventions remain duration-neutral. Though duration neutrality mitigates balance sheet interest rate risk and distinguishes these transactions from traditional asset purchases, such operations are not reserve-neutral. The requisite expansion of central bank liquidity could be seen as a form of monetary accommodation, creating a potential misalignment with an otherwise restrictive policy posture.

While representing a valuable potential addition to the financial stability toolkit for crisis intervention, the proposal primarily focuses on managing acute symptoms of dysfunction when existing market structures become severely impaired. It does not, however, propose modifications to the fundamental equilibrium mechanism by which end-investor demand for synthetic duration is met in normal times. The authors' thoughtful attention to mitigating moral hazard with Bagehot-like principles[31] further highlights its nature as a crisis-specific intervention.

Ultimately, there is a clear distinction between a tool for crisis response, which serves a critically important function, and the focus of this paper: an architectural modification designed for crisis prevention. A crisis backstop can be engineered to rely on the legal and operational elasticity inherent in an emergency.[32] A permanent market structure cannot. It must be built upon the unambiguous legal bedrock of core sovereign function, ensuring its authority is as resilient as its underlying design.

* * *

market benchmark. The investor's required futures position (w_{fut}) is a residual of their alpha-seeking cash allocation, governed by the constraint:

$$w_{cash}D_{cash} + w_{fut}D_{fut} = D_{target}$$

A large futures overhang ($w_{fut} >> 0$) can thus arise from several distinct scenarios:

1. An investor's alpha-seeking strategy, by targeting credit spreads or expressing a relative-value view on the sovereign yield curve, results in a cash portfolio with a shorter duration than desired, necessitating a separate futures position.
2. The investor's own liability profile changes, altering their fundamental D_{target}.
3. The investor minimizes active cash security holdings by design (leading to $w_{cash}D_{cash} \to 0$), sourcing duration primarily through futures to pursue leveraged strategies.

A robust policy solution must account for risks emerging from all such possibilities, not rely on a single theory that could introduce unintended fragility if other causes are dominant.

[29] Kashyap, A. K., Stein, J. C., Wallen, J. L., and Younger, J. (2025). "Treasury Market Dysfunction and the Role of the Central Bank." BPEA Conference Draft, Spring 2025. Washington: The Brookings Institution.

[30] Kashyap et al. provide a compelling quantification of the structural mismatch, noting that with pension and insurance liabilities exceeding \$25 trillion, the supply of long-duration corporate credit is less than \$3 trillion. Based on these LDI pressures, they adopt the simplifying modeling assumption that a fixed fraction (θ) of an investor's total sovereign duration position is held via futures, an allocation that is consequently "invariant to the basis." While end-investor allocations are in practice more price-sensitive—governed dynamically by the net spread after basis costs (see Section V-C)—Kashyap et al. powerfully establish that basis pressure is a permanent architectural feature of current market design rather than a transient anomaly.

[31] The authors helpfully frame the public provision of balance sheet capacity as an application of the Friedman Rule, arguing that the social cost of public balance sheet provision is negligible compared to the fragility of private leverage. Crucially, they identify a functional isomorphism where a credible commitment to cap spreads ex-post converges with policing the basis to zero ex-ante. This insight provides the conceptual foundation to operationalize that isomorphism at its source—enabling the Treasury, as primary issuer, to embed a friction-free baseline directly into market design. Direct execution at issuance resolves the friction permanently, avoiding the moral hazard and operational complexities inherent in relying on contingent, ex-post central bank interventions.

[32] Any central bank intervention in exchange-traded futures would face significant statutory constraints, as financial derivatives are not explicitly authorized for Open Market Operations under Section 14 of the Federal Reserve Act. Relying on 'incidental powers' (Section 4) or asserting functional equivalence with forward-settling obligations (analogous to the TBA MBS market) remains legally untested and complicated by exchange margin regimes. Furthermore, while emergency interventions can bypass these constraints in extremis via Section 13(3) SPVs, such emergency measures are generally seen as legally incompatible with a standing, preventative facility. This creates a degree of durable legal ambiguity ill-suited for permanent market design—contrasting sharply with primary debt issuance, which rests on plenary statutory authority (31 U.S.C. Chapter 31) explicitly dedicated to managing public debt terms.

In summary, existing and proposed measures predominantly focus on strengthening intermediaries or providing crisis tools, offering only incremental improvements to the current market architecture. Yet, they largely leave unaddressed the fundamental imbalance: end-investor synthetic duration demand met by leveraged private intermediation. Thus, these approaches risk treating market stress symptoms rather than offering a foundational solution to the underlying structural issues causing recurrent vulnerabilities.

VI. The Natural Counterparty

Structural frictions dissolve most cleanly when supply meets demand directly, in the precise form sought. In the Treasury market, that alignment rests on the sovereign's latent capacity as a producer of duration.

Market structures diverge sharply regarding the presence of natural sellers. Commodity futures (e.g., energy, metals, agriculture) feature inherent sellers: physical producers pre-selling future output to lock in forward pricing with end users. Conversely, financial futures (e.g., equity indices, currencies, SOFR) generally lack a direct producer analogue.

Treasury securities, and by extension the futures contracts that reference them, present a distinct hybrid case. They possess a critical duality: while fundamentally financial instruments, they originate from a single sovereign producer, a characteristic more aligned with physical commodity markets where forward demand connects directly to primary supply.[33] By this structural logic, the U.S. Treasury emerges as the ultimate **natural counterparty**[34] to the market's demand for synthetic duration, acting as the sole entity capable of fulfilling the underlying exposure at its origin.[35]

* * *

At present, the sovereign's natural counterparty role is outsourced to a multi-stage, leveraged intermediation chain at a significant structural cost, introducing known fault lines into the financial system.[36] Internalizing this delivery mechanism is not an expansion of mandate, but a structural optimization.[37] Rather than managing the symptoms of an outsourced bridge, such an integration represents the logical closure of the issuance lifecycle, enabling the Treasury to provide a direct conduit matching the risk-free utility of the underlying security.

VII. Closing the Architectural Gap: The Principle of Direct Sovereign Issuance

If the Treasury is indeed the natural counterparty to the market's structural demand for duration, a compelling strategic conclusion emerges: it could directly offer the forward exposure that institutional investors currently seek through less efficient, intermediated channels. This premise is the foundation of the solution proposed herein: a sovereign-led framework for **Deferred Settlement**

[33] One might initially conceive of a somewhat crude strategy: have Treasury "shelf-register" a large quantity of additional Treasury securities at each auction, intended for shorting CME Treasury futures when the basis is sufficiently wide, and for potential physical delivery against those contracts. While the practitioner's intuition might favor such direct tactical intervention in exchange-traded derivatives, the immense operational and governance challenges associated with active secondary market intervention would render such approaches unsuitable for an institution with the U.S. Treasury's mandate. In the following sections, a more formalized framework, grounded in primary issuance procedures rather than tactical market intervention, will be outlined. Nonetheless, the theoretical possibility of a sovereign-led solution serves as an encouraging starting point.

[34] While Treasury's primary mandate has historically been defined as financing the government at the lowest cost over time, the growing frequency and severity of market dysfunction originating from the demand structure for its own securities directly threatens that mandate by elevating risk premia. In this context, proactively managing the stability of its own market arguably represents not mission creep, but a direct application of its core debt management responsibility.

[35] Treasury's recent implementation of a securities buyback program serves as a critical institutional precedent. In that initiative, Treasury is proactively using its unique position as the sovereign issuer to address a secondary market challenge (liquidity in off-the-run securities). The DSA framework proposed herein is an application of that same core principle—of the issuer acting to enhance market functioning—to a primary market challenge: the architectural gap between structural synthetic duration demand and the fragile intermediary chain currently serving it.

[36] The cost is best understood as the economic carrying charge paid for passage across a private, capital-intensive bridge connecting the sovereign issuer to its end investors. Such friction manifests as a persistent premium, seen in both the cash-futures basis and negative long-end swap spreads, that sustains the entire arbitrage chain providing the intermediation service. In a more direct architecture, the bridge becomes unnecessary. The value currently paid out as a toll would instead remain within the transaction to be captured by the primary participants. Critically, direct issuance allows the Treasury to absorb the premium that investors already demonstrate they are willing to pay for synthetic duration, translating that value into lower overall yields on U.S. Treasury securities.

[37] The shift toward direct sovereign interaction is not theoretical; it is active policy. The 2026 expansion of buyback eligibility to non-dealers, formalized via the January 2026 Notice of Proposed Rulemaking (NPR) and informed by Departmental assessments that current intermediation can be "challenging and costly," validates the feasibility of bypassing dealer-centric bottlenecks. This initiative provides the foundational legal and operational rails, specifically direct bilateral agreements and the FedTrade Plus infrastructure, required for a sovereign-led issuance framework. The concurrent refresh of the primary auction infrastructure (TAAPS), with its targeted late-2026 cutover, further underscores that the issuance landscape is dynamic, providing a timely window to integrate modernized settlement logic into the sovereign's core issuance and auction infrastructure.

Auctions (DSAs).[38]

To evaluate the utility of this architecture, it is instructive to analyze the proposed mechanism first from its foundational principles as a single-period transaction before detailing its primary application as a continuous duration hedge.

A. *The Bedrock: A Sovereign Forward Contract*

Consider an asset manager seeking to lock in a core Treasury allocation, for instance, targeted 5-year duration, at a price determined today for settlement in one month. Currently, approximating this exposure requires navigating dealer-intermediated bond forwards or exchange-traded futures, both of which are tethered to the capital-intensive private intermediation chain analyzed earlier. The DSA framework is engineered to bypass these constraints, providing a sovereign-grade pathway for investors to secure the same forward exposure.

In structure, the DSA is not a new financial product, but a procedural refinement of the primary auction: **a direct, sovereign-issued forward instrument for physical delivery**. A participant secures an allocation of a specific CUSIP through the established issuance process, but the corresponding payment and delivery of the security are deferred by one month, aligning with the next issuance cycle. The deferred term provides immediate utility for investors managing forward cash flows or for liability-driven institutions seeking to align future asset deliveries with known liabilities.

While the physical delivery mechanism is a complete and sufficient utility in itself, it also serves as the structural foundation for the program's primary use case: providing sustained, synthetic duration exposure. Conceptually, an investor could maintain such exposure manually by taking delivery of their settling bond, selling it into the secondary market, and simultaneously securing a new DSA position in the next auction.[39] The section that follows details how the framework refines this manual process into a seamless, automated roll mechanism.

B. *A Seamless Roll Mechanism*

Maintaining a long-term duration overlay through manual secondary market transactions introduces cumulative operational friction, execution fees, and bid-ask slippage, mirroring the costs familiar to institutional participants navigating the quarterly Treasury futures roll. The DSA framework eliminates this overhead by offering an automated, constant-maturity roll mechanism.

Rather than taking physical delivery, an investor maintaining synthetic exposure undergoes a monthly position novation. The expiring forward contract is automatically cash-settled against its prevailing market value, and a new forward position in the incoming on-the-run security is established simultaneously.

Consider an investor who enters a June 5-year note DSA at par (100). At the July auction, the June issue trades at 98. Upon roll execution, the June contract closes at market value, and the investor remits the 2-point mark-to-market loss per 100 notional to the Treasury. Concurrently, a new DSA position is opened in the July 5-year note. The investor's duration hedge is fully restored to the on-the-run benchmark without manual secondary market execution.

The credibility of the roll relies on objective pricing. To insulate the process from administrative fiat, roll values are calculated using a transparent pricing engine derived directly from market-observed yield parameters, as detailed in Section XI.

This framework aligns instrument design with investor intent. Accommodating static, low-velocity duration through a direct sovereign channel relieves derivative markets of structural inventory pressures, freeing exchange-traded contracts to focus entirely on dynamic hedging and price discovery.

C. *Pricing Mechanics and the Basis Premium*

As a standard forward transaction, the settlement price of a Deferred Settlement Auction accounts for the time value of money to satisfy no-arbitrage conditions. The full (dirty) invoice price is adjusted to incorporate the cost of carry over the settlement cycle, benchmarked to short-term risk-free rates such as SOFR or prevailing Treasury bill yields.

In addition to short-term financing costs, the all-in price incorporates a small, positive basis premium calibrated within an initial range of 5–15 basis points annually.[40] Setting the premium within such a band ensures the

[38] The legal basis for the DSA framework is grounded in the broad authority granted to the Secretary of the Treasury under 31 U.S.C. § 3121 to set the *terms and conditions* for the issuance of government obligations. The mechanisms proposed herein are structured as an exercise of this direct debt management authority. A detailed legal analysis is presented in Section VIII.

[39] The transaction sequence is neutral to net market supply; the investor's secondary market sale is directly offset by their absorption of the new primary issuance. This is distinct from the investor's economic position, which is necessarily settled at prevailing market yields, resulting in a realized gain or loss.

[40] The calibration of the basis premium is best conceptualized as an optimal control problem, where the Treasury's objective is to manage a dynamic equilibrium: balancing a premium sufficiently compelling to drive the structural migration of addressable synthetic demand against one wide enough to prevent both an unwanted shift away from standard settlement and an outsized, price-induced expansion of synthetic-only duration strategies. Further qualitative considerations on dynamic pricing and governance levers are detailed in Sections XIII and XV.

DSA functions as an economically superior substitute for private market hedges, where synthetic duration costs have typically spanned an estimated 10–30 basis points in futures and 30–90 basis points in swaps across recent market regimes, driven by bank leverage constraints and intermediary hurdle rates.

This discrete structure is engineered specifically to serve static, strategic duration positions. In long-horizon allocations, high-velocity trading turnover, though often mistaken for true market depth, acts as an execution drag rather than genuine warehousing capacity. At the same time, maintaining a positive basis floor prevents standard cash buyers from unnaturally shifting into deferred settlement, ensuring the facility satisfies genuine synthetic demand without cannibalizing regular primary auction clearing.

The practical application of the basis premium is straightforward, added directly to the carry-adjusted forward price at settlement. For example, consider a 5-year note issued at par (100) in June under a 5.00% short-term risk-free rate regime. Assuming a 10-basis-point policy basis premium, an investor rolling or taking delivery in July settles at a dirty forward price of 100.425 per 100 of face value.[41]

D. *An Architecture of Mutual Gain*

By internalizing duration delivery, the framework generates mutually reinforcing gains across systemic stability, participant economics, and sovereign debt management:

For the financial system, the DSA mechanism enhances stability by transitioning the market away from leveraged intermediation toward robust, principal-to-principal liquidity. Shifting structural duration demand to a sovereign-backed pathway achieves a direct deleveraging of the Treasury market, mitigating March 2020-style fire-sale risks. Furthermore, this shift eases baseline repo demand across funding markets, relieving the pressures that recently prompted the Federal Reserve to launch Reserve Management Purchases.

For market participants, the transition materially improves the cost and operational profile of duration sourcing. By directly aligning the interests of the sovereign issuer and the institutional asset owner, investors gain access to synthetic duration at a cost that meaningfully undercuts private alternatives, in many cases by up to half or more. Concurrently, the seamless roll mechanism provides a resilient pathway for maintaining strategic hedges, free from the frictions and execution slippage of the quarterly futures cycle.

For the sovereign, the program converts a persistent structural friction into an immediate fiscal dividend. Instead of dissipating value across an intermediary chain, the embedded premium is retained by the Treasury. While the direct capture of a 10 basis point basis on a conservative \$500 billion adoption translates to an immediate \$500 million annual saving, this represents only the baseline layer. As detailed in Section XV, the cumulative recovery—combining basis capture, the tightening of competitive auction clearing, and the compression of systemic risk premia—is projected to generate an annual dividend exceeding \$1.7 billion.

Stemming directly from the Treasury's active stewardship of its issuance architecture, this recurring dividend transforms the DSA program into a self-funding infrastructure upgrade. Proactive modernization allows the sovereign to simultaneously enhance systemic resilience, reduce investor execution drag, and secure substantial, lasting fiscal savings for the taxpayer.

E. *From Principle to Operational Practice*

The strategic diagnosis is clear: the trillion-dollar friction in the Treasury market arises from a misalignment between how duration is issued and how it is accessed. By offering a sovereign-grade forward instrument, the DSA framework solves the alignment problem in principle, and in doing so, neutralizes the full array of systemic, policy, and fiscal vulnerabilities analyzed across the preceding sections.

Establishing the conceptual and fiscal merits of this framework is, however, only the first step. Policy innovation of such magnitude remains incomplete without a robust operational framework. The analysis must therefore pivot from the strategic *what* and *why* to the practical *how*. To bring this principle to life while safeguarding the integrity of the world's most important market, Part II delivers the complete operational blueprint, addressing the requisite initiation mechanism, legal and risk management architecture, and settlement design.

[41] The all-in dirty forward invoice price, assuming a 30-day settlement cycle under standard Actual/360 money-market convention, is:

$$100.425 = \underbrace{100}_{\text{Initial Price}} \times \left(1 + \underbrace{\frac{0.05 \times 30}{360}}_{\text{Interest Component}} + \underbrace{\frac{0.001 \times 30}{360}}_{\text{Basis Premium}}\right).$$

Part II

A Resilient and Pragmatic Implementation Framework

Transitioning the principle of direct sovereign issuance from theory to market reality requires navigating a critical dual mandate. The architecture must not only realize the economic benefits outlined in Part I: Foundations of a Sovereign Solution, but also safeguard the operational sanctity of the world's benchmark asset.

The comprehensive implementation blueprint detailed below serves as the target end-state architecture to satisfy that standard. While this design is not a monolithic, all-or-nothing proposition—offering modular, simplified graduations for early or pilot stages (detailed in Part IV)—the framework is presented here in its fully realized form, moving in strict order of execution from foundational authority to the final fiscal outcome:

- **The Legal Foundation (Section VIII)** anchors the framework not as a separate financial instrument, but as a procedural modification to primary issuance terms, derived directly from the Secretary's existing statutory authority.
- **The Initiation Mechanism (Section IX)** resolves the challenge of allocating scarce deferred settlement capacity, introducing a performance-based **DSA Multiplier** to enforce bona fide participation without disrupting the primary auction.
- **The Operational Architecture (Section X)** outlines the program's continuous risk management. It leverages a private-sector Program Administrator to execute a rigorous Tri-Party collateral regime, ensuring the sovereign balance sheet is immunized against counterparty risk.
- **The Settlement Architecture (Section XI)** details the specialized servicing mechanics. It defines the automated novation process that creates the seamless roll and the **Collateral Equity Account** used to realize gains via physical delivery.
- **Bridging the Funding Delay (Section XII)** ensures seamless fiscal integration. It demonstrates how standard Treasury bill issuance effectively manages the cash-flow latency of deferred settlement, neutralizing impact on the Treasury General Account.
- **Governance and Integrity (Section XIII)** establishes the overarching stewardship framework, codifying strict participant eligibility standards and the parameters required to guarantee the program's long-term stability.

VIII. The Legal Foundation: The Authority for Sovereign Innovation

The Treasury's sovereign debt management authority is directed toward the core mandate of minimizing long-term borrowing costs *over time*. This mission is further operationalized by the Department's standing policy to "seek continuous improvements in the auction process,"[42] a directive that provides the necessary latitude to resolve structural frictions that "inhibit efficient . . . sales" of its securities. In the modern financial system, the systemic risks inherent in the market for the government's own securities have increasingly become a primary inhibitor of such efficiency.

The events of March 2020 provided a clear and urgent signal that the structural reliance on leveraged private intermediation is not merely a technical friction, but a tangible threat to the core financing mission. While the crisis exposed the taxpayer to an acute spike in borrowing costs, the underlying fragility it revealed is chronic—distorting the transmission of monetary policy and imposing a persistent liquidity risk premium on the entire Treasury curve. The DSA framework operationalizes this mandate of continuous improvement: a well-grounded tool designed to address these compounding market vulnerabilities at their source. Indeed, both the U.S. Treasury during the WWI era and the U.K. gilt market in the late twentieth century historically demonstrated that adjusting settlement timelines to match prevailing macro pressures is a classic exercise of sovereign issuance authority.[43]

[42] U.S. Department of the Treasury (2019). "Financing the Government." Accessed August 19, 2025.

[43] Few concepts in finance are entirely novel, and the temporal flexibility of primary debt settlement is no exception. During the First World War, the U.S. Treasury departed from standard immediate settlement by offering structured installment payment plans for

To fulfill its core mandate in the face of modern systemic risks, the legal architecture for the Deferred Settlement Auction framework is grounded in the comprehensive authority granted to the Secretary of the Treasury under 31 U.S.C. Chapter 31 to manage the public debt. This well-established statutory foundation confers a dual authority, empowering the Treasury to innovate in both the substance of its securities and the procedure of their issuance.

The power to define the fundamental *substance* of an obligation, derived from 31 U.S.C. §§ 3102–3104, enabled prior modernizations such as Treasury Inflation-Protected Securities (TIPS). That substantive power is complemented by the authority to prescribe the procedural *conditions* of a sale, codified in 31 U.S.C. § 3121, which provides a strong statutory basis for modifying auction parameters.[44] Operationally, such procedural authority is exercised through the Uniform Offering Circular (UOC), located at 31 C.F.R. Part 356. By leveraging the existing administrative architecture, the Treasury can integrate deferred settlement as a standard condition of sale without necessitating new legislative action.

In effect, the entire DSA framework represents an exercise of the Treasury's sovereign debt management authority, rather than creating a new financial market for tradable instruments.[45] The authority established in the preceding paragraphs therefore logically extends to the design of the operational architecture required for the program's safe implementation. Its core components, including mechanisms to ensure bona fide participation, a framework to neutralize counterparty risk, and a process to manage the position lifecycle, are not standalone products. They are inseparable administrative controls integral to the primary issuance function itself, flowing directly from the powers vested in the Secretary by 31 U.S.C. § 3121.

The Treasury's own application of its statutory authority has proven to be dynamic rather than static. The recent securities buyback program, in particular, serves as a key interpretive precedent. It demonstrates the Treasury leveraging a long-standing power in a novel manner to proactively address a modern market structure vulnerability: poor liquidity in the secondary market.[46] The DSA framework is therefore best viewed not as a legal departure, but as an application of this same principle of adaptive stewardship, leveraging core issuance authorities to enhance the resilience of the primary market.

While the statutory foundation for the program is robust, institutional durability is best secured through broad administrative consensus. To that end, one potential path to preempt jurisdictional ambiguity and forge institutional alignment would be through inter-agency consultation. For example, a formal arrangement (such as a joint policy statement or Memorandum of Understanding with the Federal Reserve, SEC, and CFTC) could formalize the program's unique status as a tool of sovereign debt management, ensuring durable legal and operational certainty for all market participants.

* * *

The proposed architecture—grounding the solution in established statutory authority, animating it with the core sovereign mandate, and supporting it with a resilient administrative framework—is a deliberate design choice. While the cleanest theoretical path for such an innovation would be explicit legislative endorsement, the program is engineered for the pragmatic imperative of the current environment: to be fully implementable within the existing legal and regulatory landscape.

Liberty Loans, allowing retail subscribers to commit to the war effort through staged cash calls. An institutional analogue emerged in the U.K. gilt market during the late 1970s and persisted into the early 1990s. These "partly paid gilts" required only an initial fraction of the purchase price at auction, with the balance called weeks or months later. In their partly paid-up state, the securities were actively exchanged among secondary market participants, with the forward payment obligation transferring automatically to each new holder. These structures eventually faded as paper-based scrip tracking became too cumbersome and the primitive bottleneck of manual cash staging across decentralized banking networks was resolved by modern electronic payment rails. More fundamentally, as global debt markets matured, sovereign issuers standardized on plain-vanilla, regular, upfront cash auctions, outsourcing the provision of leverage and deferred exposure to highly efficient complexes of private futures, swaps, and repo. Today, the critical bottleneck is no longer the mobilization of cash, but the structural asymmetry of a heavily financialized asset-management sector whose one-sided duration demand periodically overwhelms otherwise highly functional private derivatives and funding markets, making a direct sovereign option not a nostalgic retreat, but the logical completion of modern market structure.

[44] The authority over auction procedure is broad and specific. 31 U.S.C. § 3121 empowers the Secretary to prescribe regulations on the "conditions under which the obligation will be offered for sale" (at subsection (a)(2)), the "dates for paying principal" (at (a)(5)), and "other conditions" (at (a)(7)). Critically, subsection (e) of the statute insulates such discretion from legal challenge, stating that a decision of the Secretary regarding an issuance "is final."

[45] The unique character of the DSA program distinguishes it from instruments governed by other regulatory bodies. A challenger might contend that its economic substance resembles that of a financial derivative, potentially bringing the entire program under the jurisdiction of the SEC or the CFTC. Such a view, however, overlooks the program's defining purpose: its sole function is to govern the terms of a primary sovereign issuance, not to create tradable instruments for public price discovery. This administrative purpose aligns the framework with the Treasury's direct debt management powers and established exemptions for transactions in government securities (e.g., 7 U.S.C. § 2(c)(1) regarding CEA applicability, and the 'exempted securities' status under Section 3(a)(2) of the Securities Act of 1933 and Section 3(a)(12) of the Securities Exchange Act of 1934).

[46] The buyback program is authorized under a long-standing statute, 31 U.S.C. § 3111, originally intended for standard debt redemption and management. Its modern application for strategic liquidity support confirms the Treasury's view that its mandate includes sophisticated market stewardship, not merely mechanical issuance.

The result is therefore not merely a theoretical legal justification, but an actionable blueprint designed to be executed within existing statutory boundaries, demonstrating that the Treasury possesses both the authority and a defined mechanism for implementation.

IX. The Initiation Mechanism: Solving the Allocation Challenge

The design of the DSA initiation mechanism is governed by a single operational requirement: integrating deferred settlement while preserving the absolute integrity of the primary Treasury auction. Maintaining a unified price discovery event for each new security is paramount. A separate auction for deferred-settlement bonds was analyzed and rejected; such an approach would fragment the aggregate pool of demand and undermine the status of the on-the-run security as an unambiguous global benchmark.

To enhance the primary market rather than divide it, the framework adopts an administrative model that operates as a strategic extension of the existing issuance process. This architecture is designed to establish orderly access and enforce the bona fide participation necessary to ensure the DSA program strengthens market structure without introducing new vulnerabilities.

A. *The Architectural Trilemma*

Integrating a Deferred Settlement Auction involves managing a distinct set of design trade-offs governed by three core principles. These requirements define the baseline criteria for a functional solution.

- **Auction Integrity.** The preservation of the single-price auction is the most inviolable principle. As the bedrock of sovereign financing, this mechanism is simple and transparent, eliciting honest price discovery. Its sanctity reflects a warranted institutional priority: the core systems governing the world's benchmark asset must remain inviolate.
- **Liquidity Preservation.** The on-the-run secondary market must remain deep and liquid to serve as the global benchmark. A significant portion of any new issue must therefore settle through standard channels. Consequently, a hard cap on the total notional value of DSA uptake per auction, for instance, 15–30% of the total issuance, is not a preference but a necessity.[47]
- **Bidder Certainty.** Absolute bidder certainty is essential. A participant must know, before bidding, whether a successful allocation will result in a standard or deferred settlement. Any ambiguity introduces a new risk premium, which would directly increase the Treasury's cost of borrowing.

Because participants require absolute certainty of their settlement status before pricing their bids, the allocation process cannot reside within the auction itself without compromising the price integrity the Treasury is mandated to protect.[48] This constraint necessitates that the initiation mechanism be operationally decoupled from the main auction event (see Figure 2). By design, the administrative allocation window opens concurrently with the Treasury's official auction announcement and closes on the afternoon of T-1, ensuring that finalized entitlements are communicated to participants with sufficient time to finalize their T+0 bidding strategies.

B. *The Architecture of Reputational Capital*

The DSA framework is tailored to accommodate structural, low-velocity demand rather than the tactical,

[47] While the hard cap serves as the primary safeguard for on-the-run liquidity, the framework supports two auxiliary stabilizers should fragmentation risks emerge.

First, Collateral Lending: Prior to final settlement, the Treasury could leverage the Federal Reserve's existing securities lending infrastructure (acting as fiscal agent) to make the underlying securities available to the repo market. This ensures the gross supply of collateral remains available to dealers even if net ownership is segmented.

Second, Issuance Elasticity via Capital Recycling: To prevent DSA demand from crowding out cash issuance, the Treasury can maintain the on-the-run cash tranche at a fixed, maximum volume, treating the DSA allocation as an additive 'top-up' to the auction size. The proceeds from this incremental issuance can be recycled to fund the buyback of illiquid off-the-run securities, thereby satisfying DSA demand and cleaning up dealer balance sheets without altering the sovereign's net borrowing requirement.

Distinct from these active measures, it is worth noting that fragmentation pressure is primarily a feature of the program's ramp-up phase; in equilibrium, net new migration will naturally decline once the legacy position stock is fully absorbed.

[48] Any mechanism that rations entitlements among winning bidders after the auction faces an inescapable logic trap. Following the auction, if demand for DSA Entitlements is below the program's cap, the issue is moot. If demand exceeds the cap, rationing is required, which can take two forms, both of which are architecturally unsound:

(a) *Pro-Rata Rationing.* This path presents a dilemma where both outcomes are unacceptable. It either violates the Bidder Certainty principle by forcing participants to accept capital-intensive standard settlement for a portion of their successful bids, or it violates the Auction Integrity principle by disqualifying portions of bids that were submitted at or below the clearing yield. The latter path, in particular, would incentivize distortive strategic overbidding in subsequent auctions.

(b) *Price-Priority Rationing*, which would award the capped DSA entitlements to the most aggressive bids. This approach fundamentally violates Auction Integrity by creating a joint-product auction: a bid is no longer a pure expression of duration value, but is contingent on securing a scarce settlement feature. Overbidding for the feature contaminates the aggregate demand curve and corrupts the auction's price discovery process.

Both approaches distort primary price discovery, demonstrating that entitlement allocation must precede the auction.

high-frequency requirements serviced by the futures complex. This distinction is foundational. For an institution migrating a multi-year strategic hedge, the allocation process represents a one-time strategic engagement to establish a position, not a recurring tactical exercise. Consequently, any potential delay in securing a full allocation is merely a minor operational friction, not a failure of strategy. These operating dynamics allow for an administrative allocation process that prioritizes long-term stability and rewards consistent, good-faith participation.

* * *

To preserve the deep liquidity of the on-the-run security, a prudential cap on DSA uptake is non-negotiable. The cap necessitates a mechanism to allocate the finite capacity, a function the framework resolves through a non-discretionary, pre-auction process on Day T-1. The cycle initiates with the Treasury's standard auction announcement, which specifies the total DSA capacity available and a maximum per-entity request limit. Such baseline gatekeeping ensures broad distribution and precludes market cornering.

During the allocation window, participants submit a request for a specific notional to the Program Administrator.[49] If the offering is oversubscribed, requests are filled on a weighted pro-rata basis, with each participant's requested notional scaled by their prevailing **DSA Multiplier**, a dynamic score (detailed below) reflecting the institution's history of bona fide participation. Upon the close of the T-1 window, participants receive a confirmed **DSA Entitlement**, creating a binding programmatic obligation to submit competitive bids in the primary T+0 auction for the equivalent notional.

This design transforms the multiplier from a simple rationing tool into a direct feedback loop that aligns participant incentives with the program's stability and neutralizes strategic sterilization.[50] Future access is automatically prioritized for participants who demonstrate the most proficient use of the facility: those most effective at converting entitlements into winning bids.[51] [52]

C. The Incentive Architecture

The DSA Multiplier is a dynamic governance metric that reflects a participant's objective performance history. Recalibrated after each monthly cycle, it is governed by a single input: the **Notional-Weighted Fill Rate (NWFR)**.[53] Defined as the total notional of securities won relative to the DSA Entitlements allocated to them, the NWFR measures a bidder's ability to convert entitlements into successful bids. Serving as a rolling measure of standing, the score determines the applicant's pro-rata share in oversubscribed offerings for the subsequent month, making the multiplier an asset whose value is highly sensitive to recent bidding performance.

The framework operationalizes this sensitivity through a direct update function, engineered to translate auction performance into reputational capital:

$$M_{t+1} = \min(1.0, (M_t \times \text{NWFR}) + \text{RR})$$

where M_{t+1} is the Multiplier for the subsequent cycle; M_t is the prevailing Multiplier ($M_0 = 1.0$ for new entrants); NWFR is the Notional-Weighted Fill Rate; and RR is the Regeneration Rate. In functional terms, this mechanism imposes a strict "use-it-or-lose-it" discipline on allocated capacity: an institution that secures a scarce entitlement at T-1 but fails to convert it into a successful bid at time T+0 will see its future allocation capacity automatically curtailed.

The **Regeneration Rate (RR)** is the framework's primary stabilizing parameter. It ensures a bidder's standing remains recoverable after a single month's underperformance, while making the 1.0 ceiling a status that must be actively earned through the proficient conversion of entitlements into successful bids. This architecture compels institutions to manage their eligibility standing as a finite resource, creating a

[49] The framework engages a designated private-sector fiscal agent to serve as Program Administrator, leveraging existing institutional infrastructure. The complete operational architecture is outlined in Section X.

[50] Sterilization occurs if a participant secures a scarce entitlement but fails to submit a competitive bid in the main auction, effectively wasting a portion of the program's capacity. This behavior is mitigated by the performance-based weighting system.

[51] A purely market-based T-1 auction for the entitlements themselves was analyzed and rejected. While theoretically elegant for price discovery, it would necessitate the creation of a new, operationally critical auction for a derivative-like instrument. The administrative model achieves the same anti-gaming objectives without this significant institutional friction.

[52] A system based on refundable financial deposits was discarded due to the intractable challenge of enforcement. Distinguishing a good-faith bid that narrowly misses from a bad-faith bid would require a quasi-judicial process of subjective judgments. The reputational model sidesteps this by utilizing the fill rate, an undeniable, post-auction fact, as the sole metric of performance.

[53] For any cycle in which a participant does not request a DSA Entitlement, their NWFR is programmatically set to 1.0. This ensures the mechanism penalizes only failed participation, not voluntary absence.

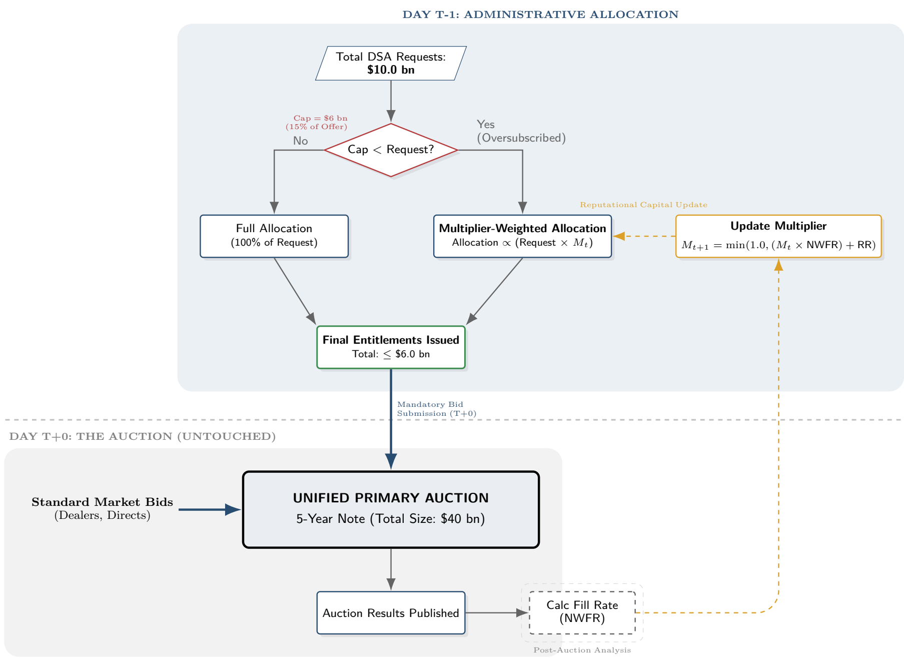

Fig. 2: Decoupled Allocation and Execution. The architecture protects the single-price discovery event on T+0 by isolating administrative rationing to a pre-auction window on T-1. This creates a closed-loop incentive structure, where post-auction performance (NWFR) is programmatically converted into reputational capital, ensuring scarce capacity is directed toward bona fide participants.

powerful incentive for bona fide bidding.[54]

Integrity is upheld by a tiered, non-discretionary enforcement architecture. A failure to submit bids for the full notional of an allocated entitlement (a *No-Show*), or a sustained period of non-competitive bidding, results in an automatic suspension. For cases of persistent non-performance, the Treasury retains the sovereign prerogative of permanent exclusion.[55] This backstop provides the final guarantee of the system's long-term reliability.

* * *

By procedurally decoupling administrative allocation on T-1 from market execution on T+0, the framework satisfies structural duration demand while leaving the core mechanics of primary price discovery untouched. Having established this foundational pillar of access and commitment, the analysis now turns to the execution architecture required to manage the position's ongoing lifecycle: the risk management protocols, settlement mechanics, and overarching governance.

X. The Operational Architecture: A Public-Private Partnership

The extended settlement architecture of the DSA framework incorporates a rigorous, dynamic margining regime to neutralize counterparty risk. While the Treasury possesses sovereign authority, attempting to replicate high-frequency collateral management capabilities within the official sector would introduce unnecessary operational overhead and development risk.

A more resilient and structurally efficient pathway utilizes the existing Financial Agency Agreement (FAA) authority

[54] The potential for the multiplier to incentivize aggressive bidding has been a central consideration in the system's design. The pressure to secure a high fill rate is real, particularly during the program's initial, high-demand phase. However, such pressure is not unbounded; it is governed by a robust and multi-layered economic equilibrium.

A sophisticated participant understands that collective DSA bidding pressure will be reflected in the auction's clearing yield, creating the risk of a *stop-through*, where the auction clears at a yield below the prevailing When-Issued rate. The resulting yield gap imposes an immediate mark-to-market (MTM) loss as the security's price normalizes toward the When-Issued mid-market post-auction. The participant's strategy is therefore anchored by a capital budgeting constraint: the immediate MTM cost of a stop-through cannot rationally exceed the franchise value of securing long-term access to the basis savings the DSA provides.

This PnL boundary is endogenously reinforced by the system's own long-term dynamics; as the DSA program successfully meets structural demand, it will naturally compress the futures basis, diminishing the very premium that justifies paying an entry cost. The resulting equilibrium is further stabilized by the fact that the target cohort of participants (Directs/Indirects) already achieves high fill rates (70–85%), indicating they bid from a position of relative strength, not desperation. Accordingly, the system is calibrated, typically via a 10–15% Regeneration Rate, to ensure bidding behavior remains within a predictable and self-limiting range.

[55] This authority includes the right to compel an orderly wind-down of a participant's outstanding DSA book, requiring the physical settlement of all positions. Such enforcement ensures the program's operational integrity cannot be compromised by a non-performing counterparty.

to engage a designated private-sector Program Administrator. This model leverages the proven, high-capacity infrastructure of the custodial banking sector to execute daily Tri-Party collateral management, margin valuation, and asset segregation, ensuring the program operates with maximum efficiency and safety from inception.

A. A Framework of Defined Roles and Authority

The integrity of the operational model rests on an unambiguous delineation of responsibilities between the policymaker and its administrative agent.

- **The U.S. Treasury as Policymaker:** The Treasury defines all governing policies and parameters for the DSA program. This includes setting the official margin methodology, determining eligible collateral and haircuts, establishing the basis pricing terms, and defining the eligibility criteria for all participants. The Treasury's role is that of the ultimate policymaker and sovereign authority, retaining full control over the initiative's design and strategic direction.
- **The Program Administrator as Operational Agent:** The Treasury would contract with a designated commercial bank, typically a Global Systemically Important Bank (G-SIB) with established Tri-Party collateral management infrastructure, to act as Program Administrator, an arrangement that applies the Treasury's standing Financial Agency authority to the specialized task of collateral management.[56] The selection would be subject to a formal, competitive process.[57] The Administrator's role is strictly that of a logistical agent, not a principal. Operating under a Master Custody and Administration Agreement, the Administrator executes the rules-based policies set by the Treasury. This includes the calculation of daily margin requirements based on Treasury pricing, the management of segregated collateral flows, and the maintenance of the ledger of record. Crucially, the Administrator does not interpose its own balance sheet or guarantee participant performance. Instead, it provides the secure, segregated infrastructure where participant collateral is pledged directly to, and held for the benefit of, the Treasury.

B. A Fully Collateralized Risk Framework

The operational architecture is engineered to achieve a single outcome: the complete protection of the taxpayer from counterparty default risk. Unlike the mutualized risk models typical of central clearing, the DSA framework utilizes a Tri-Party Collateral Management regime, ensuring that the Treasury's exposure is secured not by a mutualized guarantee fund or the balance sheet of an intermediary, but by a dedicated pool of eligible assets pledged directly to the sovereign. The Administrator executes this via a rigorous, dynamic two-part margin structure:

- **Initial Margin (IM):** Upon position entry, participants are required to post a baseline deposit of high-quality collateral. Treasury policy sets this requirement using a through-the-cycle methodology calibrated to historical stress periods (e.g., March 2020). By setting a robust static floor rather than a procyclical dynamic rate, the architecture ensures taxpayer protection without exacerbating liquidity pressures during market volatility. To ensure pristine asset quality and liquidity in stress scenarios, collateral eligibility and haircuts are strictly aligned with the Federal Reserve's Standing Repo Facility (SRF) framework, accepting U.S. Treasury securities, Agency debt, and Agency MBS—alongside U.S. dollar cash balances.[58]
- **Variation Margin (VM):** On a daily basis, the Administrator executes a mark-to-market valuation of all outstanding positions against the designated sovereign closing benchmark. Any participant whose position has incurred a loss is required to post additional Variation Margin to fully collateralize the deficit. These calls are automated and must be met within a standard T+1 settlement window.[59]

Operational integrity is enforced through strict asset segregation and established tri-party control agreements. Participant collateral is held in *bankruptcy-remote*

[56] The engagement of a private-sector entity to manage specialized infrastructure is a routine exercise of the Treasury's administrative authority. Under 12 U.S.C. §§ 90 and 265, the Secretary is broadly empowered to designate insured depository institutions as Financial Agents to perform 'reasonable duties' on behalf of the government—a power actively delegated to the Fiscal Service via Treasury Directive 16-37. This established Financial Agency Agreement (FAA) framework routinely relies on commercial banks to supply the specialized technological systems and custodial capacities not natively maintained by the Treasury, a principle of conceptual utility previously applied to the custodial functions of the myRA program (see GAO-17-176). Crucially, under 12 U.S.C. § 90, the Secretary may select these agents through an administrative designation process distinct from standard federal procurement regulations, an authority which, combined with the standing, indefinite funding mechanism of 12 U.S.C. § 5018, ensures that the program's operational requirements are met with a degree of speed and focus not possible under traditional acquisition channels.

[57] While 12 U.S.C. § 90 grants the Secretary broad discretion in the selection process, Treasury policy typically favors formal competition to ensure best value and transparency for significant Financial Agent designations.

[58] To eliminate commercial bank credit risk, cash collateral is not held as a general deposit liability of the Administrator. Instead, the framework is designed with a daily sweep into sovereign-backed vehicles, such as short-term Treasury obligations or Federal Reserve overnight facilities.

[59] While this dynamic two-part margin structure represents the gold standard for counterparty risk mitigation, the Treasury's sovereign risk tolerance may favor operational simplicity over high-frequency precision. As a robust alternative to daily VM, the framework can be calibrated to rely exclusively on a deeply stressed, static upfront margin. Because one-month Treasury drawdowns are historically bounded across tenors, a conservative buffer can virtually immunize the sovereign against default risk while rendering the position operationally dormant until the monthly settlement event.

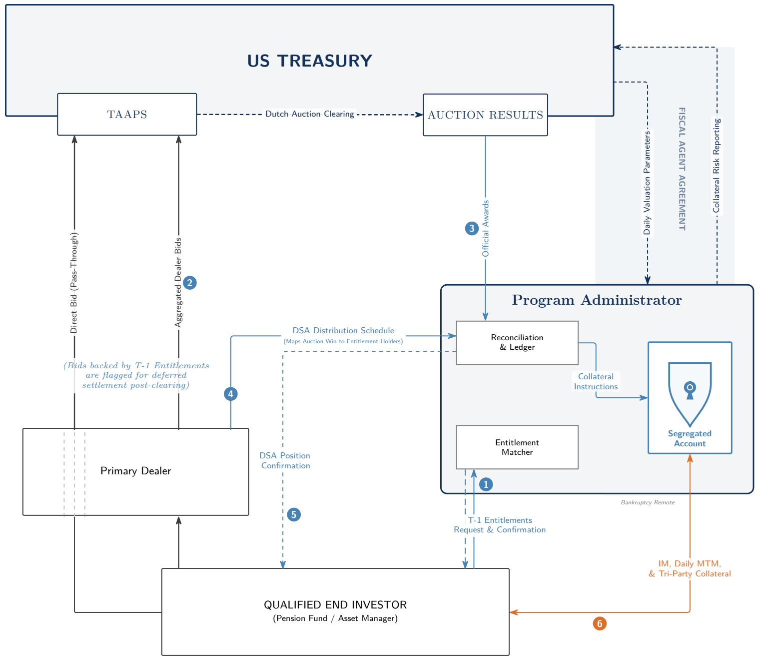

Fig. 3: The DSA Operational Architecture—A Segregated Model for Execution and Risk Management. The diagram illustrates the bifurcated workflow that separates the standard auction execution from the fully collateralized risk management of the DSA position. The operational lifecycle begins prior to the events shown (Stage 0), with a Primary Dealer verifying a Qualified End Investor and facilitating the opening of their segregated account with the Program Administrator. The subsequent transaction sequence is as follows: 1 The investor requests and receives a T-1 entitlement, establishing the administrative designation required to flag their primary auction bid for deferred settlement. 2 On auction day (T+0), the investor submits their bid via standard market channels. 3 The official auction awards are transmitted directly to the Administrator, who 4 reconciles them against the dealer's Distribution Schedule to record the final DSA position on the official ownership ledger. 5 A confirmation is sent to the investor, closing the transaction loop. 6 Finally, the position is fully collateralized via Initial Margin and daily Variation Margin, managed on an ongoing basis via a tri-party arrangement. This architecture ensures the primary auction remains untouched while the sovereign's credit risk is fully immunized.

custody accounts, legally pledged to the U.S. Treasury.[60] So structured, assets remain isolated from the credit risk of both the Administrator and the introducing intermediary, providing the Treasury with unencumbered access to collateral in the event of a default.[61]

The design of the dynamic margining regime culminates in a core principle: robust infrastructure enables operational simplicity. While equipped with the full capacity for daily, high-frequency collateral exchange, the system is designed to empower its target cohort—collateral-rich, long-horizon institutional investors—with a more efficient model, facilitating a low-maintenance approach where voluntary collateral headroom in excess of the mandatory IM allows daily mark-to-market fluctuations to be absorbed without frequent, active intervention. In practice, the system aligns its rigorous risk mechanics with the strategic, low-velocity nature of the underlying demand.

C. *The Bifurcated Access Model*

To ensure seamless market access while maintaining strict risk segregation, the framework bifurcates the functions of trade execution and collateral management. Access is contingent upon a one-time administrative requirement: every prospective end investor must establish a segregated **DSA Collateral Account** with the Program Administrator.[62] This account functions as the exclusive venue for all margin activity, mapping directly to the Administrator's official ledger of record where the participant's DSA positions are maintained and validated.

[60] While the framework utilizes the Program Administrator's high-capacity infrastructure for daily collateral management, the Secretary maintains the unilateral authority to direct the immediate transfer of pledged assets to a Federal Reserve Bank. Under 31 C.F.R. § 357.12, once a security interest is marked on the books of a Federal Reserve Bank, the United States possesses a perfected priority that supersedes all competing claims. This mechanism secures direct, unencumbered access to the collateral pool, effectively collapsing the private intermediation chain and neutralizing intermediary credit risk during systemic instability.

[61] To protect the public interest in extreme tail-risk scenarios where pledged collateral proves insufficient (e.g., a gap move exceeding the margin coverage), the framework must preserve the Treasury's right of full recourse. The Master Administration Agreement would therefore be drafted to designate any unsecured deficiency as a direct debt owed to the United States, granting it the status of a prioritized claim in any subsequent insolvency proceeding. This structure establishes that the taxpayer's claim against the general assets of the defaulting participant holds maximum legal seniority.

[62] The establishment of a DSA Collateral Account does not require the migration of an investor's broader asset base. The framework supports standard Third-Party Pledge (Tri-Party) arrangements, allowing investors to retain primary custodial relationships while pledging eligible assets to the Administrator. Operationally, this is facilitated through established industry protocols (ranging from bilateral SWIFT MT540/542 "free of payment" instructions for specific asset transfers to automated Tri-Party MT527 messaging for large-scale collateral management) ensuring seamless integration with existing institutional workflows.

The operational lifecycle is governed by three distinct phases:

The Onboarding Phase (Institutional Integration): To eliminate redundant operational overhead, the program adopts a **Delegated Reliance Model** for identity verification. Primary Dealers, serving as the interface for end investors, perform the requisite Know Your Customer (KYC) and Anti-Money Laundering (AML) checks. The Program Administrator relies on these established dealer certifications to activate the investor's DSA Collateral Account, subject to the eligibility thresholds in Section XIII. This architecture leverages the banking sector's existing compliance engine, allowing the Treasury to maintain rigorous gatekeeping without building a duplicative internal verification bureau.

The Entitlement Phase (T-1): To preserve the integrity of the Reputational Multiplier mechanism (see Section IX), the allocation of bidding rights occurs directly between the end investor and the Program Administrator. Investors submit their notional requests via its secure portal, independent of their brokerage relationships. The Administrator validates these requests against Treasury-defined limits and issues confirmed Entitlements to the investor. This ensures that the rationing of capacity is governed solely by the investor's own performance history, unclouded by intermediary aggregation.

The Execution Phase (T+0): With Entitlements secured, trade execution uses the existing primary market infrastructure, accommodating both indirect and direct access models:

- **Indirect Bidding via Primary Dealers:** The vast majority of participants will use Primary Dealers as their conduit for trade execution. The dealer aggregates client orders and submits a consolidated bid into the auction via TAAPS. Crucially, the dealer also transmits a post-auction **Distribution Schedule**, mapping the aggregate award to specific client Entitlements and instructing the immediate segregation of DSA positions from the dealer's own settlement obligations.
- **Direct Bidding:** Sophisticated institutions maintaining Direct Bidder status may submit bids directly through TAAPS, validated against their pre-confirmed Entitlements.

The Settlement Handoff: Upon the release of auction results, the Program Administrator executes an automated reconciliation, matching Treasury award notifications against Dealer Distribution Schedules and Direct Bidder logs.[63] Validated positions are booked to the DSA ledger, establishing a direct obligation between the Treasury and the end investor. Critically, the booking completes the Primary Dealer's role; because the bid is pre-mapped to a segregated collateral account, the dealer acts purely as a transactional conduit and never enters the chain of risk, title, or balance-sheet liability for the DSA position.[64]

With the program's risk neutralized via segregation and its access model defined through this bifurcated workflow, the focus can now shift to the design of the settlement mechanism itself—a design uniquely tailored to the institutional mandate of a sovereign issuer.

XI. The Settlement Architecture: A Sovereign-Aligned Framework

The DSA settlement architecture is structured for strict institutional alignment with the U.S. Treasury's core mandate as a sovereign debt manager. The resulting framework is *transactionally quiet*, achieving the economic precision of daily mark-to-market risk management without requiring the Treasury to assume the operational profile of a conventional, high-frequency derivatives counterparty. Toward this end, the framework decouples risk management from cash settlement through a four-part operational lifecycle: participant obligations are daily collateralized (Section XI-A), accrued gains are recorded via the CEA (Section XI-B), the portfolio is rolled through a spread-based pricing engine (Section XI-C), and a dedicated mechanism facilitates the realization of gains and the restoration of the strategic hedge (Section XI-D).

A. *Participant Liabilities: Real-Time Collateralization*

A core requirement of the program is that the sovereign balance sheet must be immunized against counterparty credit risk. Accordingly, the securing of participant liabilities is immediate, automated, and fully collateralized, operating through the Tri-Party margining architecture established in Section X. Daily position mark-to-market[65] variations are absorbed continuously

[63] For Direct Bidders, the Administrator reconciles Treasury award notifications directly against confirmed T-1 Entitlements, bypassing the Dealer distribution schedule.

[64] The handoff transitions the Primary Dealer's role from a capital-intensive principal to a capital-efficient agency provider. While this model shifts revenue dependency away from matched-book financing spreads, it aligns with the post-GFC regulatory imperative for balance sheet optimization. Dealers are compensated through a market-driven execution fee (analogous to the FCM model), generating a revenue stream that requires effectively zero regulatory capital allocation.

[65] Daily valuation is anchored to the indicative mid-price provided by the Federal Reserve's New Price Quote System (NPQS) 3:30 p.m. snapshot. The use of this benchmark aligns the program's risk management with the Federal Reserve's own view of the market, providing a prudent and non-discretionary source for collateral valuation.

via collateral posted to the Program Administrator rather than direct cash exchanges with the Treasury. Because investors retain the economic interest in securities pledged as Variation Margin to cover intra-month fluctuations, net mark-to-market deficits accrue a daily debit balance at SOFR. This interest component is settled in cash alongside realized price losses at each monthly novation.[66]

The resulting intra-month regime serves a dual purpose: operationally, it insulates the Treasury from daily cash flows while maintaining real-time credit protection; legally, it enforces the daily valuation discipline required to satisfy institutional regulatory and tax accounting standards.[67]

B. *Sovereign Liabilities: The Collateral Equity Account*

In contrast to participant liabilities, which are secured via the external posting of collateral, sovereign obligations arising from participant gains are managed through the **Collateral Equity Account (CEA)**. The CEA is not a distinct custody account but a dedicated book-entry record maintained by the Program Administrator. It functions as an operationally quiet administrative ledger, tracking the net equity accrued to the participant's position without necessitating daily cash outflows from the Treasury.

The CEA is activated only when the prevailing market price of the security exceeds the participant's effective entry or novation price. While price movements below this threshold govern the calling or release of VM, moves above the threshold are recorded as unrealized accrued equity. To ensure economic parity with daily cash-settled instruments, and to prevent structural value leakage, the CEA balance earns interest daily at the applicable overnight benchmark rate (typically SOFR). Consequently, the CEA balance provides the economic equivalent of a daily cash-settled gain, with the full balance carried forward at each monthly novation to preserve the position's cumulative unrealized gains.

While the CEA exists as a book-entry record, its value is not merely notational; it represents a tangible financial asset. The framework provides a direct mechanism for participants to realize this accrued equity by converting it into physical Treasury securities, a process detailed in Section XI-D. Furthermore, the CEA balance serves as sovereign-grade internal collateral, directly offsetting mandated Initial Margin. This creates a ladder of progressive collateral reduction: as the CEA balance grows, external collateral requirements diminish dollar-for-dollar. Once the accrued equity surpasses the mandated IM threshold, the participant may withdraw all external assets, leaving the position fully secured by the Treasury's own accrued obligation.

By obviating the need for daily cash payments from the sovereign, the settlement architecture bridges issuer efficiency with participant compatibility. The mechanism provides the Treasury with the economic precision of a daily margin cycle while effectively neutralizing its pre-novation participant-facing footprint, operating in complete harmony with the long-horizon mandates of its target institutional audience.[68]

C. *The Monthly Novation: A Sovereign-Aligned Roll*

As established in Part I, the seamless roll is the DSA program's primary use case, designed to provide investors with continuous, constant-maturity duration exposure. Operationally, the daily mark-to-market and collateralization process detailed above culminates in this central monthly novation—timed to coincide with the corresponding auction (see Figure 4)—where the open position is formally rolled by exchanging the outgoing obligation for a new forward position in the incoming on-the-run security.

The programmatic integrity of a seamless roll is predicated on a robust pricing methodology designed to withstand market volatility and technical friction. The architecture that follows establishes a settlement price using a transparent, market-based process designed for resilience against manipulation. Furthermore, the system must do so with a level of operational simplicity befitting a sovereign issuer and within a structure that is institutionally and legally sound.[69]

The framework achieves these objectives by anchoring the entire transaction to the single most robust and unimpeachable pricing source available: the clearing price of the concurrent primary auction itself. Indexing both the entry and exit legs of the transaction primarily to this single pricing event ensures stability, minimizes ambiguity, and insulates the roll from extraneous market

[66] This intra-month interest component is recorded through the Collateral Equity Account (CEA) described in Section XI-B. Any negative CEA balance is required to be fully cash-settled back to zero at each monthly novation.

[67] This mandatory daily valuation and collateralization regime establishes the "system of marking to market" required under 26 U.S.C. § 1256(g)(1)(A). By doing so, it aligns the instrument's tax character with that of regulated futures contracts, creating the functional parity necessary for broad institutional adoption while preserving the instrument's status as a direct sovereign obligation.

[68] The framework satisfies the corresponding requirement in 26 U.S.C. § 1256(g)(1)(A), i.e., that margined amounts "may be withdrawn," through a dual mechanism rooted in the Collateral Equity Account. First, as the CEA balance grows, it directly substitutes for Initial Margin, permitting the participant to withdraw their previously pledged external collateral. Second, the participant has the right to realize the full value of the CEA by taking physical delivery of securities, as detailed in Section XI-D.

[69] Simpler alternatives were analyzed and rejected as inadequate. A manual roll (requiring the investor to take delivery, sell the bond, and re-enter the T-1 subscription process) fails on the grounds of operational friction. An iterative extension of the original bond's delivery date fails to meet the core constant-duration objective. A purpose-built, market-based solution is therefore not a preference, but a necessity.

Fig. 4: From Intra-Month Risk Isolation to Physical Novation. The framework maintains a passive sovereign profile by buffering fluctuations through the dual mechanisms of margining and the CEA, ensuring the Treasury's only active cash-flow engagement is the receipt of final cash payments to clear realized participant losses at the month-end settlement event.

noise.

The novation is executed through a two-legged process:

- **The Entry Leg**. The new DSA position is initiated at a level derived directly from the clearing yield of the main Treasury auction, aligning it perfectly with the single, official price for that security.
- **The Exit Leg**. The protocol avoids the ambiguity of sourcing an independent market price for the outgoing security. Instead, it adjusts the auction clearing yield by the market-observed yield spread between the outgoing (previously on-the-run) and the new When-Issued security.[70] This spread is derived from the yield equivalents of the Volume-Weighted Average Prices (VWAP)[71] of both securities over a designated 30-minute window,[72] beginning shortly after the auction results are released.[73]

Operationally, the execution of these two legs is formalized as a procedural novation of the underlying physical delivery obligation.[74] [75] To execute the roll without absorbing competitive auction supply, the participant's reinvestment into the incoming DSA is structured as a non-competitive, add-on award, an approach aligning cleanly with established sovereign debt management authorities.[76]

[70] This spread adjustment is a necessary calibration to ensure economic neutrality. It accounts for the idiosyncratic yield profile of the maturing vintage (specifically its transition to off-the-run status and its one-month duration roll-down), ensuring the exit price represents a precise fair-market valuation for that specific CUSIP.

[71] The framework leverages the standard, high-integrity data infrastructure that already underpins the U.S. Treasury market. The primary source is FINRA's Trade Reporting and Compliance Engine (TRACE), validated in real-time by firm quote streams from major interdealer platforms. This ensures the VWAP calculation is robust, transparent, and derived from the regulatory record.

[72] By initiating the observation window (e.g., 1:15 PM to 1:45 PM ET) after the concentrated volatility of the auction release has normalized, active trading in both vintages is guaranteed to be fully underway. This window typically occupies a news-neutral phase of the trading day, providing a robust, high-integrity environment for the spread calculation.

[73] To be institutionally viable, the pricing mechanism must be resilient to adversarial stress. The designated 30-minute VWAP window represents a theoretical surface for exploitation where a market actor might attempt to distort the yield spread between the outgoing on-the-run security and the new When-Issued benchmark. The DSA framework is designed with a series of mutually reinforcing provisions to neutralize this risk.

The primary line of defense is inherent to the *market's own microstructure*: because these two securities are structurally the most actively traded instruments at that tenor, any attempt to sustain non-economic pricing during a period of such heightened market scrutiny would be met with deep, opposing liquidity from the global fixed-income relative value community, ensuring that the cost of fighting market gravity remains punishingly high.

This is reinforced by a secondary tier of *procedural safeguards*: the authority to randomize observation intervals within a wider communicated block—or extend the window to a full-day VWAP across the When-Issued period—prevents precise execution targeting. Furthermore, as noted in Footnote 47, leveraging Federal Reserve securities lending infrastructure supports secondary market borrow availability, tempering localized supply squeezes during the roll window.

Should market dislocation persist, the framework incorporates an *administrative fallback*: indexing the roll spread to an augmented yield curve spline. Because a fitted curve draws upon a broad cross-section of sovereign issues, it naturally filters out idiosyncratic, single-CUSIP pricing distortions.

Finally, operational integrity is maintained through *active surveillance and adaptive calibration*. Operating in coordination with FINRA, the Treasury monitors TRACE data for anomalous execution patterns. If empirical monitoring identifies recurrent execution friction or structural spread distortions, the Treasury retains the authority to recalibrate the program's baseline basis premium for future cycles, ensuring the sovereign's fiscal dividend remains protected.

[74] The novation executes as a net cash settlement of the mark-to-market variance prior to physical delivery—a mechanism conceptually analogous to a standing buyback and re-issuance. The participant's commitment to acquire the outgoing issue is extinguished and replaced by the incoming forward position, avoiding gross operational DvP churn over Fedwire.

[75] While roll pricing is locked on auction day (T+0), the resulting cash variance settlement and collateral adjustments occur on the security's formal issue date.

[76] This mechanism draws upon the Secretary's broad statutory authority to define issuance terms and extinguish debt obligations

The framework's internal logic is most clearly demonstrated in the case of reopening auctions. Where the outgoing and incoming securities share a CUSIP, the roll is executed as a pure price novation. The forward price is reset to the new clearing yield, resulting in a perfect mechanical offset. This allows for a frictionless roll that maintains the structural link to the physical market, bypassing the computational requirement of a spread-based adjustment.

Anchoring execution to the primary auction through transparent, rules-based pricing establishes clear price integrity across the roll cycle. By maintaining these operational standards, the framework satisfies the criteria for a Qualified Board or Exchange (QBE) designation.[77] Such a designation secures tax parity for participants while firmly maintaining the program's status as a direct sovereign obligation, exempt from the regulatory complexities of organized exchanges.

* * *

The monthly novation concludes with a simultaneous Cash-for-Collateral exchange, releasing pledged Variation Margin upon receipt of the participant's cash settlement for net price losses and accrued intra-month debit interest. This consolidated mechanism minimizes the sovereign's operational interface to a single monthly touchpoint, providing full credit protection without burdening daily debt management with intraday cash maintenance.[78]

(31 U.S.C. §§ 3111, 3121). Characterizing the roll as a procedural novation of a physical delivery obligation preserves the instrument's status as a primary government debt transaction under 7 U.S.C. § 2(c)(1). Furthermore, processing reinvestments as non-competitive add-on awards prevents rolling volumes from reducing the supply available to competitive bidders in the main auction, preserving the price discovery engine of the primary market.

[77] To establish tax parity for taxable participants and prevent the creation of inefficient tax deferral structures, it is proposed that the Secretary designate the DSA program as a Qualified Board or Exchange (QBE) under 26 U.S.C. § 1256(g)(7)(C). A QBE status would draw upon the functional principles established in Revenue Ruling 87-43, which recognized trading arrangements governed by adequate rules as qualifying for § 1256 treatment even if execution occurs outside a traditional exchange floor. Crucially, the legal architecture is engineered so the resulting tax classification remains legally distinct from, and does not confer status as, an "organized exchange" under the Commodity Exchange Act. The program falls outside the statutory definition of an "organized exchange" under 7 U.S.C. § 1a(37) by virtue of two deliberate structural constraints: first, all execution occurs on a strict principal-to-principal basis; and second, the governance framework limits disciplinary sanctions exclusively to the suspension or exclusion of participants. This architectural integrity keeps transactions unambiguously within the scope of the government securities exclusion codified at 7 U.S.C. § 2(c)(1).

[78] In the interest of institutional clarity, these monthly cash flows are best characterized not as independent gains or losses, but as 'Settlement Value Adjustments.' This framing reflects the economic reality that such payments are merely the explicit, accelerated recognition of the mark-to-market valuation changes inherently impacting the Treasury's entire debt stock. By settling the variance at the monthly novation, the Treasury is simply bringing forward a component of the interest-expense adjustment otherwise realized over the security's full maturity.

D. Realizing Gains & Restoring Notional

While the novation cycle provides a definitive settlement point for intra-month mark-to-market liabilities, the treatment of accrued investor gains is engineered to reinforce the program's identity as a tool for physical debt acquisition. Accordingly, the Collateral Equity Account functions as a legally binding claim on the sovereign ledger rather than an ephemeral accounting entry. Operating under the objective of minimizing daily cash flows, the framework provides participants with a flexible, user-initiated mechanism for value realization: the **Adjusted Invoice Settlement**.

This mechanism allows account holders to monetize their accrued equity without requiring the Treasury to disburse cash. During any monthly novation cycle, a participant may elect to take physical delivery of a portion of their position. In such an exchange, the investor's CEA balance functions as valid tender, applied dollar-for-dollar to offset the cash cost of the securities.[79] The physical delivery requirement is a deliberate structural feature; by precluding immediate cash monetization, it renders the instrument intrinsically unsuitable for high-velocity tactical trading and reserves its utility for long-horizon strategic asset allocation.

The logical culmination of the principle is the special case where an investor can elect to settle a notional amount precisely equal to their CEA balance, allowing them to extract a fully paid-for Treasury security in a transaction that is cash-flow neutral from the Treasury's perspective. Because the CEA balance represents a previously recognized financial gain, its application to the settlement price functions as a payment, not a discount. Consequently, the security is delivered with a standard cost basis equal to its full market value, preventing the creation of Original Issue Discount (OID) complexities.[80]

Settling via Adjusted Invoice, by definition, reduces the notional value of the investor's ongoing hedge.[81] The

[79] While an acquisition of securities to redeem CEA balances might superficially appear forced, it is an operational non-event for most Qualified Participants. Because QPs are the market's anchor cash buyers, they already purchase Treasuries at auction as a matter of routine. The Adjusted Invoice enables them to fund a portion of those existing purchase obligations with accrued equity instead of cash.

[80] Under 26 U.S.C. § 1256, the gain represented by the CEA balance is marked-to-market and recognized for tax purposes. Therefore, applying this balance to acquire the security is economically equivalent to a reinvestment of fully realized, taxable capital. The Issue Price of the delivered security remains the full Auction Price, rendering it fungible with standard CUSIPs.

[81] For example, consider an investor with a $500 million DSA position (notional) and a cumulative $5 million CEA balance. The investor elects to take physical delivery of $5 million in face value, applying their CEA balance to offset the prevailing market cost of those securities. This transaction effectively exercises and extinguishes the $5 million notional slice of their forward position. To prevent the investor's aggregate duration hedge from shrinking to $495 million, the restoration mechanism automatically generates

program counteracts such notional attrition with an integrated solution for participants requiring constant notional exposure: **Automatic Notional Restoration**.

An investor electing to settle via Adjusted Invoice is entitled to a non-competitive, add-on allocation for a new DSA position in the concurrent auction, equal to the notional amount they extracted. This mechanism, functioning as a synthetic analog to SOMA reinvestments, restores the participant's strategic hedge to its original size.[82] The result is a closed-loop system where gains are realized as physical assets, and the structural hedge is maintained without friction.

E. *Finality via Physical Delivery*

The mandatory settlement method for any terminated DSA position—one not rolled forward in the subsequent novation cycle—is the physical delivery of the underlying Treasury security against payment of the all-in forward price. By precluding purely cash-settled terminations and requiring all accrued equity to be realized through sovereign debt acquisition, the framework anchors the instrument directly in the physical market. This operational finality reinforces the instrument's legal standing as a primary debt offering rather than a vehicle for financial speculation.

XII. Bridging the Funding Delay: Treasury Bills and the DSA Mechanism

Consolidating the operational foundations developed thus far, Deferred Settlement Auctions promise a rare alignment of interests: asset managers gain a more efficient tool for duration management; the Treasury secures fiscal savings; and systemic risk is reduced by connecting the sovereign issuer directly to the end user through a professionally managed architecture. The viability of this comprehensive framework, however, hinges on addressing one final operational imperative: bridging the cash-flow timing gap inherent to deferred settlement.

A. *Treasury Liquidity Management*

The Treasury issues public debt to bridge the gap between fiscal receipts and expenditures. While this remains the core principle, for established sovereign issuers, a significant portion of bond issuance is used to refinance existing debt. Regardless of issuance purpose, the Treasury anticipates a corresponding increase in its General Account for planned disbursements. Deferred settlement, however, would introduce a timing variance into the receipt of auction proceeds.

When an investor secures bonds at auction in June, with payment deferred to July, receipt of settlement proceeds is effectively deferred by a month. At modest scale, the delay is inconsequential. However, the program is expressly designed for substantial volume, accommodating structural duration delivery preferences and alleviating repo market pressure, not marginal adjustments.

The timing discrepancy is exacerbated by the ongoing roll feature, potentially deferring final settlement for quarters or years post-auction. Consequently, effective Treasury fund receipt becomes contingent on investor-initiated settlement, rendering it inherently unpredictable for fiscal planning. This investor-driven timing is incompatible with the requirement for predictable, forward-looking fiscal management.

B. *The T-Bill Bridge: A Substitution, Not an Addition*

The optimal tool for managing cash-flow latency resides within the Treasury's existing operational toolkit: Treasury bill issuance. Rather than creating new administrative machinery, the Treasury can bridge the timing gap by adjusting its weekly bill supply.[83] What appears to be a significant fiscal deferral is thus absorbed as a routine calibration across the high-frequency bill schedule, requiring zero expansion of the Treasury's operational footprint.[84]

One might initially worry that issuing hundreds of billions in additional bills would exert upward pressure on short-term yields, potentially diminishing the program's fiscal dividend. However, a flow-of-funds analysis shows that deferred settlement does not increase the net demand for money market financing; it re-routes it. Resolving the apparent paradox requires tracing the flow of funds to its origin in the short-term funding complex.

Under the prevailing architecture, a significant portion of Treasury bond issuance is already funded by money market cash lenders (see Figure 1). This liquidity currently flows through a private intermediation layer, where hedge fund arbitrageurs must secure repo financing to warehouse these positions. Consequently, the basis

a new $5 million notional DSA position in the current auction.

[82] By design, this add-on allocation is for a new forward-settling DSA, not an outright security, ensuring the entire hedge restoration process is self-contained within the program and has no direct, contemporaneous impact on secondary market supply or trading volumes.

[83] The cost to Treasury for this bridge financing is effectively zero, as it recovers the interest expense through the short-term rate accrual embedded in the DSA pricing mechanism (see Section VII-C).

[84] Operationally, distributing the monthly DSA requirement across the deep weekly bill schedule prevents localized supply shocks. This approach adheres to the Treasury's 'regular and predictable' issuance framework, avoiding reliance on ad-hoc financing vehicles. Any minor day-count mismatches between bill settlements and DSA obligations are routinely absorbed by normal cash-balance fluctuations in the Treasury General Account (TGA).

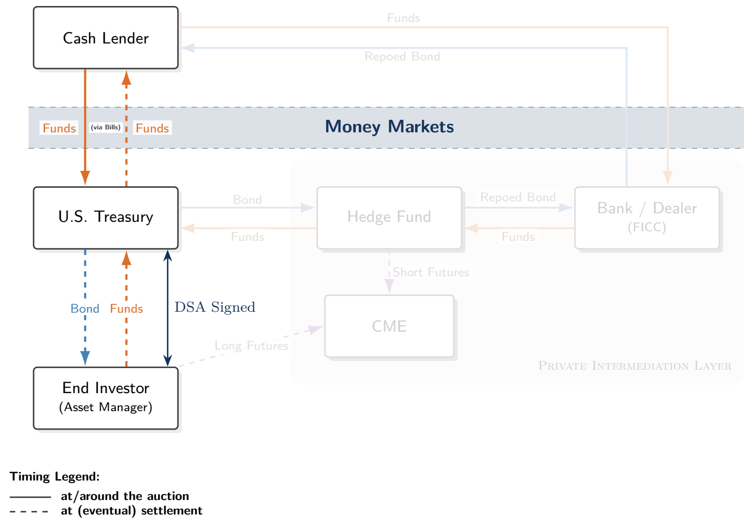

Fig. 5: Rerouting the Funding Circuit under Deferred Settlement. In contrast to the intermediated chain in Figure 1, the DSA framework directly connects sovereign supply to institutional demand. Treasury bill issuance bridges the cash-flow latency by sourcing liquidity directly from money market lenders, substituting private repo borrowing dollar-for-dollar. In the process, private intermediaries are relieved of warehousing static sovereign duration, eliminating a primary source of systemic fragility while leaving aggregate financing demand unchanged.

trade exerts an inherent, if indirect, upward push on short-term rates. As such, short-term funding pressure represents an incumbent friction of the current market structure, rather than a net-new risk introduced by the DSA framework.

As end investors migrate their duration demand to the DSA channel (as depicted in Figure 5), the resulting issuance of Treasury bills to bridge settlement gaps[85] is concurrently offset by a corresponding decline in repo borrowing from withdrawing hedge funds. The net effect is a *substitution* of borrowing demand, not an addition. The ultimate lenders are merely swapping counterparties—from a repo-funded arbitrageur to the U.S. Treasury itself. With T-bills and repo satisfying the same core mandate for high-quality liquid assets, the aggregate financing extended to support the holding of Treasury securities is unaffected.[86]

Mechanically, the shift achieves a compositional rebalancing of the public debt stock. By replacing a portion of physical coupon supply with synthetic duration and bill financing, the Treasury preserves its aggregate duration and interest rate profile intact, while directly reducing the volume of securities requiring secondary market intermediation.

The ensuing transition ultimately enhances the resilience of sovereign funding. By replacing complex, leveraged repo chains with direct sovereign borrowing, the Treasury internalizes the financing of its own duration. The result is a more robust money market ecosystem, where a private funding chain prone to stress-induced dislocations is superseded by the unparalleled depth and operational simplicity of the U.S. Treasury bill market.

XIII. A Framework for Governance and Integrity

With the operational architecture established, the focus shifts to safeguarding its long-term integrity. The success of the program requires a governance regime built on aligned incentives rather than rigid prescription. The design that follows provides not a static set of rules, but a flexible system of policy levers, enabling the DSA program to function as a responsive instrument of sovereign debt management, capable of continuous calibration against empirical data and evolving market dynamics.

A. *The Program's Structural Role*

The DSA framework is governed by the principle of Functional Complementarity. It acknowledges the U.S. Treasury futures market as the indispensable anchor of global price discovery, a venue whose nearly $1 trillion in average daily notional volume provides the necessary

[85] This substitution creates no conflict with the statutory debt limit. Because official debt is incurred on the settlement date, the bridge T-bill would be the only instrument counted until the DSA bond settles. Upon settlement, the bond replaces the T-bill, leaving the total debt subject to the limit unaffected.

[86] The near-perfect substitution is evidenced by 4-week T-bills commonly trading within a tight 2 bps band of overnight tri-party repo rates under stable rate expectations and minimal basis-trade interference. As a direct sovereign obligation with a deep secondary market, T-bills are almost uniquely insulated from issuance-related pricing concessions; their relative pricing is shaped mostly by Fed-managed system liquidity.

While deep, UST repo markets lack that same degree of quantity-related issuance inelasticity, as the Dallas Fed's research earlier showed in Section I. In other words, there is no cleaner vehicle than bills to absorb extra supply without shifting yields at the short end. To the extent that the basis trade's repo borrowing has already indirectly pulled the bill curve upward, transitioning this exposure to direct T-bill issuance would not aggravate the issue.

depth for tactical risk transfer and high-frequency hedging.

The demand addressed by deferred settlement represents a structurally distinct, low-velocity segment of the interest rate complex. Such flow is the non-discretionary residual of strategic duration management, concentrated in the mechanical execution of the quarterly roll, defining a recurring exchange between asset owners maintaining a strategic hedge and the arbitrageurs who currently warehouse the offsetting basis exposure.

Deferred Settlement Auctions seek a structural migration of demand rather than the elimination of the basis trade.[87] By absorbing the weight of passive, long-term positioning, the program purifies the futures market's price discovery mechanism, allowing it to better reflect the fair-value exchange of tactical risk. This surgical intervention targets the static duration that drives leverage-induced fragility, while leaving a residual basis as a natural, and necessary, indicator of idiosyncratic liquidity and tactical demand.[88]

B. Defining Participant Eligibility and Enforcement

To ensure the program serves its intended purpose as a tool for strategic duration management, participation would be governed by a transparent **Qualified Participant Standard**, establishing a universal eligibility threshold based on institutional scale, as measured by a defined minimum Assets Under Management (AUM) floor. Crucially, it restricts participation to specific, verifiable institutional structures.[89] The use of objective legal classifications provides a transparent gatekeeping mechanism, reserving issuance capacity specifically for the constituency of natural asset owners.

To protect the sovereign balance sheet, enforcement mandates are designed for **automatic risk neutralization**. A failure to meet a margin call or bidding commitment therefore results in the immediate termination of the position and liquidation of pledged collateral.[90] While this primary risk-control function is absolute, the consequences for participants are tiered. A first-time failure results in a temporary suspension of bidding eligibility—an automated, time-limited safeguard against transient operational glitches—whereas a demonstrated pattern of non-compliance triggers permanent exclusion from the program. Such a governance model provides the necessary firmness to protect the Treasury against counterparty credit risk without compromising operational pragmatism.

C. Aligning with Long-Term Horizons

Beyond eligibility gates, the design relies on the inherent physics of the instrument to foster the desired institutional behavior. The DSA is engineered with specific structural features that naturally align with long-term asset allocation, rendering the instrument economically inefficient for tactical trading:[91]

- **Non-Transferability:** DSA positions are contractually non-transferable and cannot be assigned to other parties. This creates a closed ecosystem that precludes the development of a secondary market, ensuring the instrument is used solely for balance sheet holding rather than high-velocity turnover.[92]
- **Discrete Liquidity Cycles:** Unlike futures or cash Treasuries which offer continuous intraday liquidity,

[87] This migration does not obsolesce the arbitrage function; it right-sizes it. The active convergence mechanism connecting cash and futures remains a critical systemic artery, efficiently maintained by private intermediation. The DSA simply relieves this circuit of the static, capital-intensive load that currently threatens to exhaust it, ensuring dealer and arbitrageur capacity remains robust and available for the dynamic enforcement of relative value.

[88] The direct impact on exchange volume is notably limited. For every \$500 billion in migrated notional, approximately \$8 trillion in annualized execution notional is removed, accounting for the individually cleared legs of the quarterly roll for both the end investor and the intermediary. This represents roughly 4% of the complex's nearly \$200 trillion in annual volume. While such a transition would necessarily be reflected in lower baseline open interest, the high-velocity flow that constitutes true market depth remains largely intact.

[89] To ensure the program's capacity remains dedicated to the priority demographic of structural asset owners, eligibility is governed by an objective determination of legal status. Qualified Participants are limited to entities meeting one of the following regulatory or statutory classifications: (1) Registered Investment Companies (entities registered under the Investment Company Act of 1940); (2) Qualified Retirement Plans (subject to ERISA or Governmental Plans as defined under 26 U.S.C. § 414(d)); (3) Regulated Insurance Companies (subject to state insurance authority and NAIC reporting standards); (4) Tax-Exempt Organizations (as defined under 26 U.S.C. § 501(c)(3)); (5) Sovereign Entities (qualifying for exemption under 26 U.S.C. § 892); and (6) Foreign Equivalents (satisfying comparable criteria, e.g., UCITS funds or entities meeting the statutory definition of a Qualified Foreign Pension Fund under 26 U.S.C. § 897(l)). For assets managed via discretionary separate accounts, eligibility requires both the ultimate beneficial owner and the investment adviser of record for the account to independently qualify as a Participant under the criteria above.

[90] The Program Administrator functions as the executing entity for risk mitigation. Upon a participant's failure to satisfy a margin call, the Administrator is authorized to trigger the immediate termination of the position and the liquidation of collateral. This automated sequence, executed directly at the custodial level, ensures the taxpayer is protected by a definitive and rapid conversion of pledged assets into cash-settlement.

[91] This structural friction serves as the ultimate gatekeeper. For those of us who have managed high-velocity tactical strategies, the sacrifice of capital velocity and intraday liquidity is a non-starter. This ensures the program naturally self-selects for its target demographic, reinforcing the administrative eligibility criteria through the structural mechanics of the trade.

[92] While the one-month interval between settlement windows for a non-transferable position might theoretically suggest an illiquidity premium, such a requirement is largely neutralized by the operational profile of the target participant. For a liability-driven institution with asset allocations that remain ~90% static across quarters, a 21-business-day settlement cycle represents a negligible friction. Moreover, because a DSA position is an unfunded forward, it does not "lock up" principal required for withdrawals or reallocations. Participants requiring mid-month tactical flexibility can readily adopt a core-satellite construction, satisfying 80% of their requirement via the cost-efficient DSA channel while retaining a 20% residual in Treasury futures for immediate liquidation or delta adjustment.

DSA positions can only be established or adjusted during the Treasury's monthly auction cycle. This operational latency effectively insulates the program from strategies seeking to capitalize on intraday price fluctuations. Crucially, discrete settlement does not constrain a participant's ability to manage risk; rather, it reinforces a clear separation of intent, where intra-month duration shifts are executed via the high-velocity futures complex, leaving the core DSA position to serve as a stable, long-term strategic anchor.

- **Mandatory Physical Settlement:** As detailed in Section XI, the inability to cash-settle gains mandates the final conversion of the position into physical debt, ensuring that the instrument remains anchored to the economics of long-term asset acquisition rather than pure price arbitrage.

D. *Overall Program Parameters and Stability Tools*

The framework operates within high-level parameters designed to preserve systemic stability and protect the integrity of the primary market.

- **Aggregate Program Size:** A prudential ceiling on the total notional volume of outstanding deferred settlement obligations would be established, calibrated against end-investor demand and indexed to a fraction of total marketable debt (e.g., an initial cap of approximately \$1 trillion).
- **Maximum Uptake Per Auction:** To preserve robust price discovery in the cash market, the share of any single auction available for deferred settlement would be capped (e.g., 15–30%), balancing DSA demand with the need for on-the-run liquidity. Such a cap would not be overly restrictive, potentially accommodating up to \$750 billion in annual uptake under current issuance parameters.[93]
- **Exit and Termination Protocols:** To balance an indefinite term structure with sovereign control, the framework would incorporate a clear hierarchy of exit mechanisms:
  - **Participant Exit (Physical):** Investors may terminate their forward position during any monthly cycle. Final settlement is executed via physical delivery, scheduled to coincide with the incoming auction.
  - **Sovereign Call Authority:** While the instrument is open-ended for the investor, the offering's design acknowledges the Treasury's unilateral authority to terminate the program or specific vintages with reasonable notice. This *Sovereign Call* is designed to ensure the Treasury is not locked into a perpetual liability structure and allows for the orderly wind-down of the facility if debt management priorities shift. In such an event, outstanding positions would be settled via mandatory physical delivery.
- **Passive Continuation Timeline:** To reflect the indefinite nature of the instrument, the decision timeline defaults to continuity. Investors are required to provide notification only if they intend to terminate (take delivery), with such notice due no later than T-2 of the subsequent auction cycle. To safeguard primary auction price discovery, aggregate roll and termination volumes are embargoed until auction clearing. Releasing these data alongside official results maintains information symmetry regarding the conversion of synthetic positions into physical float, preventing pre-auction supply distortions.

[93] Initial scaling could be accelerated by allowing direct buyback participants to convert off-the-run sales into non-competitive DSA allocations at subsequent auctions, or through dedicated buyback-to-DSA exchange operations. This provides an additive channel for migrating legacy duration overlays that operates entirely outside the standard auction cap.

Part III

Architectural Validation & Economic Impact

With the operational blueprint in place, the analysis now turns to validating the architecture against its core stability mandates, quantifying its fiscal dividend, and mapping the transition toward a more capital-efficient market equilibrium.

XIV. Architectural Validation

To ensure the blueprint provides an operable design that satisfies its systemic goals, rather than merely an abstract proposal, the framework is tested against the design criteria defined in Part I.

- **Deleveraging the System:** The current market relies on hedge funds to act as a bridge, requiring substantial gross leverage to facilitate a straightforward duration transfer. Deferred Settlement supersedes the levered bridge with an unlevered, principal-to-principal obligation. By shifting duration delivery from the derivatives complex to the direct issuance channel, the framework fundamentally reduces the system's aggregate leverage. Concurrently, the associated contraction in repo demand stabilizes overnight rates by alleviating the pressures of inelastic financing. As a result, monetary policy transmission becomes insulated from private balance-sheet frictions, mitigating the need for reactive central bank interventions and smoothing the path toward long-term balance sheet normalization.
- **Preserving Price Discovery:** A critical constraint was the protection of the futures market's signal value. By defining DSAs as a tool for *static* holding (enforced through non-transferability and discrete liquidity cycles), the design preserves tactical price discovery within the futures complex, while the distortive weight of passive positioning is migrated to a more appropriate venue.[94]
- **Sustaining Alpha-Seeking Capital:** The framework preserves the yield-reducing benefits of institutional spread-asset allocation. By providing a resilient, direct channel for the requisite duration, DSAs allow asset managers to maintain overweight allocations to Agency MBS and corporate credit, sustaining downward pressure on those spreads, without the contingent liability of a leveraged intermediation chain. Because DSAs merely re-route existing structural demand to the issuer at source, aggregate absorption of sovereign duration remains unchanged, ensuring the transition does not exert upward pressure on baseline Treasury yields.
- **Adoption Incentives:** The solution does not rely on regulatory mandates to force participation, but on a compelling economic incentive. Beyond the inherent savings of basis compression, DSAs eliminate the persistent drag of the quarterly futures roll. By replacing high-frequency manual execution—and its associated explicit commissions, bid-ask spreads, and execution slippage[95]—with a passive, algorithmic reference update, the instrument offers a materially superior total cost of ownership for long-term duration portfolios.

By internalizing a function currently outsourced to leveraged intermediaries, the architecture achieves its primary stability mandate while generating a cascade of economic benefits: lower hedging costs for investors and, ultimately, a direct fiscal dividend for the sovereign.

XV. The Fiscal Dividend

With the structural integrity of the framework secured, the analysis now shifts to its economic outcomes. As established in Part I, internalizing the duration-delivery function allows the sovereign to reclaim the economic value currently lost to intermediary friction. The following

[94] The objection that narrowing the basis removes a valuable signal of market stress conflates a symptom with a useful diagnostic. A structural dislocation, such as a persistent, leverage-dependent basis, is a signal of architectural insufficiency, not a feature of a healthy market. A resilient design prioritizes the eradication of the underlying friction; the resulting stability is, in itself, the most informative signal of a well-functioning system.

[95] For a well-executed \$1 billion position in 5-year Treasury futures, the performance drag from rolling via the quarterly calendar spread, once averaged over a full market cycle, tends to settle around 3–8 bps annually—a figure driven less by explicit fees than by the implicit friction of intermediating predictable institutional flows. During periods of market turbulence, this cost can readily multiply several-fold as liquidity provision evaporates, with costs scaling non-linearly for larger positions and varying across tenors.

sections move beyond this theoretical premise to quantify that value, demonstrating how the architecture translates a systemic inefficiency into a recurring fiscal dividend.

The resulting gains operate across three vectors. The capture of the direct basis premium is perhaps the most intuitive; however, it is the latent intrinsic benefits, derived from realignment of the auction's competitive clearing and the compression of systemic risk, that account for the preponderance of the program's realized fiscal value. Though the empirical isolation of these effects presents a higher analytical hurdle, the fiscal savings they generate are no less real.

Crucially, while the framework requires a modest, one-time investment in its technological and administrative infrastructure, it is engineered to be rapidly self-funding. The recurring fiscal dividend generated from the basis premium alone is projected to fully amortize this initial outlay early in its operational lifecycle, making it accretive thereafter and requiring no long-term net appropriation.[96]

A. *Quantifying the Direct Dividend*

To quantify the potential savings, we model a conservative steady-state scenario. The analysis is anchored on two key assumptions:

1) **Program Adoption:** Of the more than $1 trillion in structural synthetic net duration demand, we assume a stable adoption of **$500 billion**. This is not a forecast of total demand, but a prudent estimate of the passive, long-horizon component most suitable for the DSA framework. The figure reflects a modeled market equilibrium where DSA adoption is assumed to compress the basis spread, limiting further migration once the costs of the two channels converge for the marginal user. The estimate represents the expected steady-state attractor[97] where the program successfully satisfies the market's core non-tactical demand while accounting for institutional inertia and the indispensable role of futures for tactical hedging.
2) **Retained Value:** We assume the Treasury prices the DSA to retain a basis of **10 basis points**. That level represents the lower bound of the typical 10–30 basis points p.a. cost of intermediated futures. Such a calibration ensures DSAs provide a compelling economic advantage to end investors while still generating a significant return for the sovereign.

Based on these deliberately conservative inputs, the framework is projected to generate a **Direct Dividend** of **$500 million** in annual interest expense savings. Assuming program adoption scales in proportion to the current real value of the marketable debt stock, we treat this dividend as a real cash flow. Discounted over a ten-year horizon using the 10-year TIPS real yield—the sovereign's real cost of capital—the stream produces a Net Present Value (NPV) of approximately **$4.5 billion** in capitalized market structure efficiency.

The potential for friction recovery extends beyond the futures complex. The interest rate swap market represents a significant secondary frontier; pension funds and insurers hold net receiving positions in the hundreds of billions of dollars, paying a structural premium for synthetic duration that has become a normalized, if inefficient, feature of the broader interest rate complex. While regulatory mandates may limit wholesale migration, DSAs offer a capital-efficient alternative for the *generic* duration component of these portfolios. In this segment, the largest absolute recovery of friction is possible, with potential savings that could eventually rival or exceed those derived from the basis trade alone.

B. *Capturing the Competitive Yield Recovery*

The transition from a high-cost private basis to a sovereign channel creates a significant economic surplus. If the private market requires, say, a 20 basis point friction to bridge the gap between cash and futures, and the Treasury's administrative fee is set at 10 basis points, a 10 basis point cushion remains. The capture of this surplus, whether retained by the investor or recovered by the issuer, is ultimately determined by the incentive architecture of the Dutch auction.

Under the current regime, the marginal price-setter in the auction is frequently a leveraged arbitrageur who is structurally anchored to the relative-value parity vis-à-vis the futures market.[98] This reflexive link constrains their

[96] The framework's cost structure explicitly minimizes direct sovereign expenditure. By leveraging the existing high-capacity infrastructure of a private-sector Program Administrator for the resource-intensive functions of collateral management and ledgering, the Treasury avoids significant upfront technology build costs. Instead, the Program Administrator functions under a standard service model, amortizing its own platform adaptation costs over the life of the contract. Consequently, the Treasury's implementation scope is largely restricted to legal finalization, internal accounting updates, and the automated ingestion of post-auction award data—a "read-only" integration that leaves the core TAAPS bidding engine untouched. Even accounting for the significant outlays typical of federal systems integration and legal finalization, the projected $500 million in annual direct savings ensures the program is self-funding in months, even prior to accounting for broader yield-compression benefits.

[97] The attractor logic assumes an inverse relationship between the DSA's embedded basis and adoption volume. A higher basis increases per-unit revenue but limits migration from private channels; conversely, a lower basis maximizes adoption but reduces the per-unit dividend. This trade-off ensures the aggregate fiscal dividend tends to cluster within a central band, as the product of price and volume is constrained by the market's price elasticity of demand for synthetic duration.

[98] This role is not limited to basis-oriented hedge funds. The marginal bidder may also be a bank dealer warehousing a specific CUSIP in anticipation of its entry into the deliverable basket as a Cheapest-to-Deliver (CTD) or liquid near-CTD security, holding the position until it can be efficiently swapped into a private basis trade.

TABLE I: Empirical Estimates of Yield Sensitivity ($\partial y/\partial Q$)

Study	Mkt.	Methodology	Economic Signal	Est. (bps/$1bn)
Mustafi (2024); Droste (2021)	US	HFI / Intraday	Transient Liquidity Noise	0.08
Albuquerque et al. (2024)	PT	Marginal Elasticity	Structural Elasticity Proxy	0.09
Cavaleri et al. (2025)	CH	Demand Elasticity	Structural Elasticity Proxy	0.14
Bank of Canada (2023)	CA	Direct Estimation	Sovereign Peer Benchmark	0.22
Hortaçsu et al. (2017)	US	Bidder Surplus	Implied U.S. Schedule	0.38
Bi et al. (2025)	US	HFI / Structural	Path-Dependent Repricing	0.40
NY Fed (2025)	US	Price Impact	Immediate Liquidity Cost	0.44
Treasury ODM Survey (2016)	US	Dealer Consensus	Sustained Supply Shift	1.00

capacity to lead price discovery in the cash market, as their bidding logic remains disciplined by the cost of hedging the resulting basis position against a fluctuating derivatives complex. Consequently, the arbitrageur acts as a key marginal actor, often exerting a disproportionate influence on the price-setting margin of the demand curve that defines the final clearing yield. The DSA participant, however, enters with a distinct strategic advantage. Having bypassed the arbitrage circuit, their indifference price is tied to their liability mandate rather than the futures price. They possess the latitude to bid more aggressively than the intermediary while still achieving a net duration cost significantly lower than that of the futures market.

The realization of the yield compression operates through two distinct phases. During the initial implementation, the framework creates a concentrated demand-side pressure. The prudential cap (see Section IX) transforms DSA capacity into a scarce resource, compelling participants to manage their allocation risk by bidding aggressively. Because the DSA provides a significant cost advantage over private alternatives, these participants can economically justify sharing a larger portion of their pricing cushion with the sovereign, tightening the bid stack without exceeding their indifference threshold.[99]

Once the program reaches its steady state, the benefit shifts from a demand shock to a permanent competitive supply compression. In that state, the rolling DSA book functions as a stable demand anchor—effectively a quasi-non-competitive bid.[100] For the Treasury, the dynamic represents a structural reduction in the volume of securities that must be issued to the price-sensitive, competitive market.

* * *

To quantify the terminal impact, we model the 5-year Treasury note, the median anchor for synthetic long-horizon duration demand. As of early 2026, institutional long positions in the 5-year futures complex (FV) total approximately $380 billion in gross notional value. Assuming a 50% migration to the DSA channel, and normalizing for the duration characteristics of the underlying cash securities,[101] this represents a re-absorption of approximately $206.5 billion in static duration demand.

In a steady state, the projected $206.5 billion DSA book represents roughly 4.9% of the $4.2 trillion 5-year float. Against monthly issuance of $70 billion, this book effectively displaces $3.4 billion from every auction, clearing the allocations outside the competitive bid stack. The fiscal gain is determined by the sensitivity of the stop-out yield to a given reduction in competitive supply:

$$\Delta y_{\text{stop}} \approx \frac{\partial y_{\text{stop}}}{\partial Q} \cdot \Delta Q \,.$$

Absent public bid-level data,[102] we derive this sensitivity by synthesizing a spectrum of empirical evidence[103]—from High-Frequency Identification (HFI) studies, secondary market price impact, sovereign peer analysis to dealer surveys—normalized for the 2026 U.S. 5-year market. These range from transient liquidity shocks to sustained

[99] The scarcity effect is reinforced by the fact that a lower-cost instrument naturally expands the pool of prospective bidders. By improving the net economics of 'credit-plus-synthetic' portfolios, the DSA activates demand from institutional managers for whom high private basis costs previously rendered duration-neutral credit strategies unattractive. This increased competition for the program's finite capacity ensures the sovereign captures the bulk of the structural cushion, as participants utilize their bidding latitude to secure an allocation within the competitive mechanics of the Dutch auction.

[100] While modeled as a stable feature of the auction, this participation remains fundamentally discretionary. Participants are not surrendering strategic freedom; should yields compress beyond fundamental levels, they retain the full flexibility to adjust their asset allocation or exit the DSA framework entirely.

[101] Utilizing a standard conversion factor (CF) of 0.92, calibrated to a representative 4% coupon to reflect prevailing 5-year par yields. This normalization is required to bridge the gap between the futures contract's notional and the cash market's price-sensitivity (DV01), ensuring the modeled displacement reflects the true economic volume of duration being migrated.

[102] While independent analysis is necessarily constrained by the use of public proxies, the Bureau of the Fiscal Service possesses the definitive bidding archives required to calibrate this sensitivity with absolute precision. Such internal visibility allows the sovereign to empirically validate the demand-curve slope and the resulting fiscal dividend prior to a full-scale rollout.

[103] Conceptually, we treat a reduction in auction supply and an increase in structural demand as a functional equivalence; in the closed-loop environment of the primary auction, these represent the identical realignment of the market-clearing equilibrium.

structural repricing (see Table I).[104]

To account for the facility's permanent footprint, while maintaining a prudent margin of safety, we adopt a baseline slope of 0.25 basis points per $1 billion. The chosen value represents a rigorous middle ground: it sits above the noise of transient liquidity but remains roughly 75% lower than the impact predicted by the primary dealer consensus for sustained supply shifts. Applied to the $3.4 billion steady-state displacement, this sensitivity yields a baseline yield compression of 0.86 basis points.

This compression generates a profound fiscal multiplier. Because the Treasury utilizes a single-price Dutch auction format, the yield reduction achieved at the stop-out point governs the coupon for the entire issue, extending the benefit far beyond the specific notional of the DSA book. For the 5-year note, a 0.86 basis point reduction generates approximately $361 million in annual interest savings. Scaled across the curve, where 5-year notes represent roughly 30% of the aggregate basis trade and serve as a representative median for the respective elasticities, the aggregate competitive recovery is projected to exceed $1.2 billion annually. By re-aligning the auction's priority of bids, the Treasury effectively internalizes the "New Issue Premium" that is currently dissipated through leveraged private intermediation.

Together with the direct basis capture established previously, the framework's total internal fiscal recovery is projected to reach $1.7 billion annually. This yields a 10-year Net Present Value of approximately $15 billion, providing the Treasury with a substantial and recurring dividend from the architectural modernization of its issuance process.

[104] The empirical challenge in quantifying yield sensitivity lies in distinguishing between transient liquidity noise and the structural elasticity required for a permanent shift in issuance patterns. Estimates based on High-Frequency Identification (HFI), such as Mustafi (2024) and Droste et al. (2021), which utilize intraday price changes and bid-to-cover ratios, typically capture the market's immediate response to supply imbalances and define the lower bound of sensitivity. Studies of sovereign peers with lower liquidity depth, such as Cavaleri et al. (2025) in Switzerland and Albuquerque et al. (2024) in Portugal, provide cross-market proxies that cluster around 0.14 bp when normalized for the U.S. 5-year market depth. Utilizing a proprietary dataset of U.S. Treasury bidder surplus, Hortaçsu et al. (2017) derive a slope that, adjusted for the convexity of the demand curve and projected 2026 depth, stands at 0.38 bp (moderated from an unadjusted 0.48 bp). As a definitive upper bound for one-time supply absorption, the NY Fed's trade-impact model, measured during the quietest market phases and grossed up from its $100 million baseline via the square root law, suggests a ceiling of 0.44 bp. However, the elasticity of a one-time liquidity shock is historically distinct from the permanent supply displacement inherent in the DSA framework. Research into sustained shifts, such as Bi et al. (2025), and institutional benchmarks like the U.S. Treasury's 2016 Dealer Survey, which explicitly queried dealers on the pricing of a sustained, one year increase in issuance, indicate that long-horizon sensitivities are significantly higher (often by a factor of 3x to 5x) reaching or exceeding 1.00 bp.

C. *Reclaiming Sovereign Risk Premia*

Beyond the direct and competitive dividends established in the preceding sections, the framework's ultimate fiscal resonance resides in the compression of the liquidity risk premium across the entire Treasury yield curve. In modern asset pricing, a security's value is determined by its *Stochastic Discount Factor* (SDF)—specifically, its capacity to hold its premium value and provide frictionless exit precisely when the marginal utility of wealth spikes.[105]

By decoupling duration delivery from the leveraged basis trade, the Treasury reduces the conditional covariance between market stress and forced sale dislocations. In doing so, it dampens the "wrong-way" correlation observed in recent crises, where the secondary market value of sovereign debt depreciates in concert with risk assets due to intermediary deleveraging, and restores the security's fundamental utility as a pure risk-off hedge.

While a 1 basis point compression is used here as a conservative lower bound, empirical evidence from recent market dislocations suggests that the latent liquidity risk premium embedded in the current architecture—the yield premium demanded to absorb forced-sale risks during stress—may be several multiples larger. Even at this floor, a 1 basis point reduction across the approximately $23 trillion in marketable coupon debt would begin to flow through as a recurring **Structural Dividend** of approximately **$2.3 billion** annually.

* * *

Direct sovereign issuance closes the architectural gap between duration creation and institutional absorption. By bypassing the private intermediation chain, the Treasury recaptures the economic value currently dissipated as an arbitrage toll, while end investors eliminate the persistent execution drag of rolling derivatives. Ultimately, the realignment demonstrates that systemic market resilience and sovereign fiscal efficiency are not competing choices, but joint outcomes of a modernized issuance structure.

XVI. Market Evolution, Capital Efficiency and Systemic Realignment

The current architecture incentivizes the private sector to commit scarce, high-cost risk capital to intermediate

[105] Academic framing aside, the transmission runs directly through real-world risk models. In standard mean-variance and tail-risk frameworks, any breakdown in flight-to-safety behavior erodes an asset's diversification value. As risk engines react to positive stock-bond correlations, institutional mandates systematically scale back Treasury allocations, mechanically driving up sovereign borrowing costs.

a risk-free, static sovereign flow. Such intermediation represents a fundamental inefficiency in the management of the financial system's aggregate balance sheet. By internalizing duration delivery, the sovereign releases substantial private sector capacity, both dealer balance sheet and hedge fund risk capital, allowing it to be redeployed toward higher-value functions: liquidity provision, tactical price discovery, and active capital formation.

While the transition implies a corresponding evolution in the revenue composition of the Treasury ecosystem, compressing the specific revenue pools associated with the passive warehousing of $500 billion in structural demand, it should be viewed as an architectural optimization. For arbitrageurs, it signals a transition away from the gross-notional footprint of leveraged carry strategies, characterized by a profile of consistent accrual and conditional stability; for dealers, it facilitates a shift from balance-sheet-intensive financing to capital-efficient agency execution.

Critically, the migration of these high-volume, low-margin trades eases the pressure on SLR and G-SIB limits, allowing participants to pivot scarce capacity toward higher-ROE activities. Such re-pricing of intermediated alternatives is an expected market outcome where the sovereign can directly close a structural gap at a lower all-in cost. It represents a rationalization of the capital structure, where the issuer satisfies the market's baseline duration demand, while private-sector balance sheets are released to focus on the high-velocity risk transfer where their comparative advantage is highest.

* * *

One further dimension warrants attention here, one that the current regulatory debate has begun to trace, but whose natural conclusion extends considerably further than it has yet reached.

The analytical focus on leveraged intermediaries as the locus of systemic risk reflects where the fragility is visible during transmission. It does not capture where it originates. Asset managers' structural allocation toward shorter-duration spread assets generates the duration deficit—and thus the persistent futures richness—that makes the arbitrage attractive. Concurrently, money market funds housed overwhelmingly within the same institutional complexes supply the short-term repo financing required to fund the trade at scale.

The leveraged intermediary occupies the space between these two capital allocations under the same management umbrella, performing the intermediation the asset manager's own balance sheet, for regulatory and institutional reasons, cannot. It takes on the leverage, the regulatory scrutiny, and the crisis amplification. The economic position necessitating the arrangement sits elsewhere.

Under stress, the logic of the arrangement inverts. Money market vehicles, carrying near risk-free classifications and daily-redeemable liabilities, are both entitled and frequently compelled to withdraw from repo markets when conditions deteriorate. That withdrawal forces rapid deleveraging in the intermediary layer: Treasuries are sold, futures are repurchased, the basis widens. The synthetic duration hedges supporting the spread-seeking allocation become materially more expensive at the precise moment of maximum institutional stress. In a sufficiently acute scenario, the overlay may deteriorate past the point of economic viability, prompting a retreat to cash Treasuries funded by the liquidation of the spread assets the strategy was designed to hold, producing procyclical selling in credit markets alongside the prior sovereign fire sale.

Macroprudential analysis mapping this full chain will arrive at the end-investor perimeter. Haircut requirements on cash lenders, enhanced liquidity mandates on money market vehicles, capital charges on synthetic overlays: these are the instruments of a regulatory response designed to trace fragility to its source rather than its conduit. The intermediary bears that scrutiny today. There is no inherent reason it does so permanently.

DSAs dissolve the underlying dependency before that perimeter completes its natural trajectory. Duration acquired directly from the sovereign—fully collateralized, requiring no passage through leveraged repo—carries none of the systemic interdependency underpinning this regulatory rationale. The spread-seeking allocation survives intact; only the delivery mechanism for its requisite duration hedge is modernized. A framework of this design offers end investors a considerably more tractable adaptation than the one a fully extended regulatory perimeter would require.

Part IV

The Path to Adoption

XVII. A Prudent Path Forward: A Pilot Program

The systemic vulnerabilities inherent in the existing landscape are not hypothetical, but a recurring, well-documented liability. Yet the transition to a more resilient architecture must not itself introduce fragility. A disciplined risk management framework demands real-world verification alongside conceptual soundness, making a phased pilot program the necessary bridge from blueprint to implementation.

The most viable operational model anchors the pilot to the standard three-month reopening cycle of benchmark securities. To illustrate, a DSA position established at a February 10-year new-issue auction would be rolled forward using the results from the March and April reopenings of that same CUSIP. By strictly utilizing the reopening cycle, a single-CUSIP structure provides a clean-room environment for testing the core mechanics, allowing for the precise calibration of the allocation and collateral infrastructure before layering on the broader valuation dynamics of a cross-security roll.[106]

Instead of requiring a single, high-stakes leap, the framework decomposes into a few modular dials that can be turned independently. Table II maps these choices, serving both as a summary of the program's moving parts and a pragmatic guide to scaling them. Because each stage represents a stable, secure equilibrium, the Treasury can turn these dials as volume dictates, or permanently maintain the system at a lower-complexity level. Starting with a reopening pilot represents a highly practical, low-overhead starting point.

* * *

Such a phased rollout approach creates a clear strategic proposition for the market's core asset owners. The pilot offers a low-risk opportunity to quantify the economic benefits of DSA basis and roll mechanics against existing instruments and to integrate internal workflows ahead of a full-scale launch. Participation in this phase serves as an operational proof of concept,[107] providing the technical and empirical justification required for a permanent program.[108] As such, the pilot functions as the catalyst for the partnership needed to deliver a modernized and more resilient financial architecture.

Specifically, the pilot provides the controlled setting required to achieve three primary objectives before scaling:

- **Validate the Core Mechanics:** A pilot, capped at a modest notional size (e.g., \$15–25 billion), delivers real-world validation of the program's key innovations, specifically the passive continuation (roll) mechanism and the associated netting and settlement protocols.
- **Gather Empirical Data:** Participation by sophisticated institutions provides the empirical data required to inform the optimal calibration of demand elasticity, pricing spreads, and collateral haircuts for a full-scale rollout.
- **Assess Operational Readiness:** The program serves as a concrete stress-test of the connectivity between the Treasury, the Program Administrator, and Primary Dealers acting as the designated client

[106] To further minimize operational friction during the pilot, the Treasury could implement a simplified collateral regime. Rather than launching the dynamic daily margining engine immediately, the pilot could use a static, upfront collateral deposit (e.g., 105% of notional in T-Bills). Such a model is a direct extension of the Treasury's existing risk tolerance, where the 20-year bond's auction-to-settlement cycle already creates an uncollateralized forward exposure of 6 to 9 business days. The fact that Treasury currently manages the existing risk without collateral provides a strong justification for a pilot that utilizes a much safer, fully over-collateralized structure.

[107] To ensure the integrity of the initial launch, the pilot's eligibility criteria could, in addition to the Qualified Participant Standard, utilize the same duration-adjusted auction-award metric currently employed as a screen for direct buyback access (\$35 billion in 10-year equivalents over a six-month period). While long-term eligibility may expand to broader AUM-based standards, this primary market activity threshold serves as a rigorous, pilot-specific proxy for institutional scale and technical readiness.

[108] While the pilot establishes the operational baseline, participation metrics should be interpreted as a floor rather than a projection of steady-state demand. By restricting the scope to the three-month reopening cycle, the pilot requires participants to manage the terminal liquidity of their positions at the end of each vintage, rather than utilizing the seamless cross-security roll of the permanent program. The objective is the modular validation of the settlement and collateralization plumbing, preceding a full-curve rollout.

TABLE II: DSA Implementation Roadmap: Pilot to Steady State

Operational Module	Pilot	Intermediate	Target Steady-State
1. Execution & Allocation	Non-competitive pro-rata allotment of capped capacity to onboarded participants at the clearing price.	A pro-rata subscription window open on T-1, with oversubscriptions scaled across all Qualified Participants.	Decoupled T-1 allocation managed via dynamic performance multipliers updated systematically after each cycle.
2. Risk Management	Upfront 105% collateral in bills/bonds pledged directly to the Treasury's account at the Federal Reserve under OC 7.	Standard initial margin paired with periodic, non-daily variation margin settlement to reduce operational overhead.	Daily tri-party margin with securities collateral pledged in-place and cash variation margin swept into sovereign-backed cash vehicles.
3. Position Novation & Pricing	Single-CUSIP rolls restricted to reopening auctions, requiring only a mechanical price novation to the new clearing yield.	Cross-security rolls priced using a conservative (20th pctl) yield spread derived from NPQS snapshots over the settlement cycle.	Real-time roll pricing executed via a synchronized 30-minute post-auction TRACE VWAP spread.
4. Realization of Gains	Accrued equity applied exclusively to offset the final purchase price of the security upon terminal delivery.	Monthly net cash settlement of accrued gains and losses executed concurrently with each novation date.	Accrued equity earns SOFR, offsets upfront margin, and physical delivery via equity offsets triggers non-competitive notional restoration.

interface, ensuring the execution infrastructure is robust before scaling to aggregate market volumes.

Ultimately, this validation phase provides the empirical foundation required for an informed policy decision. Absent such a defined pathway, the market remains tethered to a known source of fragility, committing the system to a permanent posture of crisis response. The pilot thus transforms an abstract proposal into a manageable project, offering a route to modernization that avoids the operational risks of a sudden structural overhaul.

XVIII. Toward a Resilient Sovereign Market

The most durable market solutions arise when perennial frictions are treated not as permanent liabilities to be managed, but as structural inefficiencies waiting to be resolved. In the Treasury market, the trillion-dollar demand for synthetic duration has always had a natural counterparty perfectly positioned to satisfy it. What has been missing is not the authority, nor the underlying infrastructure—but an actionable blueprint.

The systemic risks embedded in the Treasury basis trade are not an intractable feature of modern finance, but a signal. They indicate that institutional demand of this magnitude has outgrown the capacity of an intermediation chain reliant on unprecedented repo leverage to operate safely, revealing an architectural gap that only the sovereign, as issuer of first resort, is constitutionally equipped to fill.

In Deferred Settlement Auctions, the Treasury finds a direct operational answer to that signal. By establishing a direct conduit between the primary issuer and the institutional end user, the DSA framework internalizes the delivery of synthetic duration, replacing a capital-intensive, outsourced bridge with a fully collateralized, sovereign-grade pathway. This represents the logical completion of the issuance lifecycle: a transition from the reactive management of market symptoms to the stewardship of its own issuance architecture.

Translating the architecture into practice requires no conceptual leap, only the purposeful application of established operational rails. The blueprint detailed here leverages existing infrastructure: TAAPS, the Financial Agency Agreement framework, and proven Tri-Party collateral mechanics. The implementation cost is deliberately modest. The projected return, converting a persistent systemic friction into a direct fiscal saving of over one billion dollars annually for the taxpayer, is not.

The Treasury can continue to manage a leveraged intermediation chain whose fragility it neither controls nor can afford to ignore, or it can resolve the structural asymmetry directly—capturing the embedded basis premium to finance the government at the lowest cost over time, unencumbering monetary policy transmission, and mitigating systemic risk. The architecture is now drawn. The authority has always existed. What remains is an exercise of sovereign prerogative.

In doing so, the Treasury does more than refine an auction process; it ensures that, rather than being defended through intervention, the world's risk-free benchmark remains resilient by design.